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# An automated computational framework to construct printability maps for additively manufactured metal alloys

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In metal additive manufacturing (AM), processing parameters can affect the probability of macroscopic defect formation (lack-of-fusion, keyholing, balling), which can, in turn, jeopardize the final product's integrity. A printability map classifies regions in the processing space where an alloy can be printed with or without porosity defects. However, the creation of these printability maps is resource-intensive. Previous efforts to generate printability maps have required single-track experiments on pre-alloyed powder, limiting the utilization of these printability maps for the high-throughput design of printable alloys. We address these challenges in the case of Laser Powder Bed Fusion AM (L-PBF-AM) by introducing a fully computational, predictive approach to create printability maps for arbitrary alloys. Our framework uses physics-based thermal models and a variety of defect formation criteria. We benchmark the predictive ability of the proposed framework against literature data for the following commonly printed alloys: 316 Stainless Steel, Inconel 718, Ti-6Al-4V, AF96, and Ni-5Nb. Furthermore, we deploy the framework on NiTi-based Shape Memory Alloys (SMAs) as a case study. We scrutinize the accuracy of various sets of defect criteria and use these accuracy measurements to create an uncertainty-aware probabilistic framework capable of predicting the printability maps of arbitrary alloys. This framework has the potential to guide alloy designers to potentially easy-to-print alloys, enabling the co-design of high-performing printable alloys.

Additive manufacturing (AM) technologies have demonstrated their value across various applications, including aerospace, defense, automotive, and biomedical industries[1,2]. Nevertheless, many processing parameters (laser power, scan speed, powder layer thickness etc.) lead to a high likelihood of unsuccessful, porosity-ridden prints. The porosity ultimately undermines the structural integrity of the part and impedes the broader use of AM as a manufacturing route for critical components. In metal laser powder bed fusion AM (L-PBF-AM) processes, common defects that form include lack-of-fusion, balling, and keyhole-induced porosity.

To fabricate porosity-free parts using L-PBF-AM, it is crucial to comprehend the correlation between defects and both processing parameters and material properties. Processing parameters include laser power, scan speed, hatch spacing, substrate thickness, and more. Relevant material properties include density, transformation temperatures, heat absorptivity, thermal conductivity, and heat capacity. Printability maps are used to better understand the interplay of these material properties, processing parameters, and defect occurrence[2–8].

In previous work, experimental data played a role in printability analyses. Defects were classified based on single-track experiments and experimentally calibrated thermal models. However, the cost and time required to obtain high-quality experimental data limit the use of printability maps in high-throughput (HTP) approaches for AM materials and process co-design. Consequently, a fully automated computational framework with reasonable accuracy is essential to guide AM design efficiently in vast composition and process spaces.

In this work, we present a computational framework that generates printability maps for arbitrary alloys in an automated HTP manner. Using only nominal chemistry as an input, the proposed framework estimates material properties using CALPHAD-based databases and empirical/analytical equations. These material properties are then inputted into thermal models. Single-track simulations are performed for multiple materials and processing parameters. From these simulations, single-track melt pool dimensions are obtained. single-tracks are classified as defective or defect-free using a set of printability criteria, which are functions of melt pool

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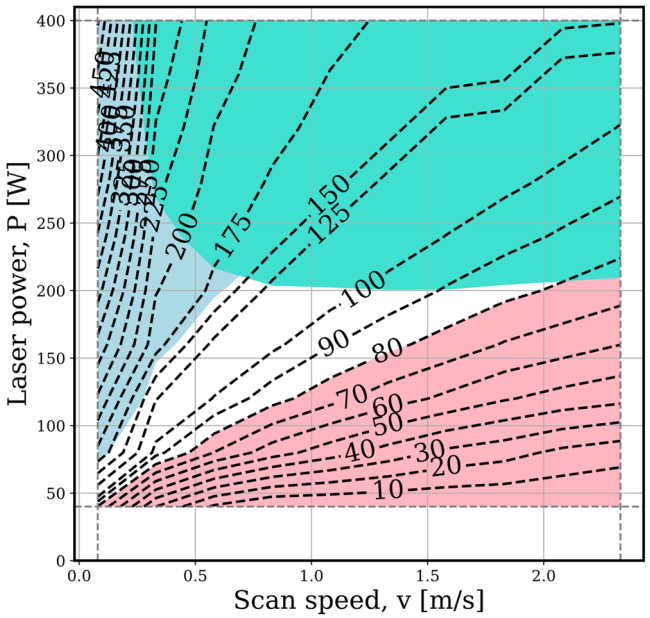



**Fig. 1 | Example of a printability map.** The figure displays an example printability map, which illustrates various L-PBF processing conditions linked to fabrication defects. The map employs colors and bounding lines to denote defect regimes for lack-of-fusion (pink area), balling (green area), and keyholing (blue area). In addition, the dashed lines indicate the maximum hatch spacing needed to avoid insufficient overlap between tracks. The scan speed has units of meters per second (m/s) and the laser power has units of wattage (W).

dimensions and thermophysical properties. These printability criteria identify power-velocity combinations that result in different defect modes.

As a case study, we apply the framework to commonly fabricated AM alloys: 316 Stainless Steel, Ti-6Al-4V, Inconel 718, AF96, and Ni-5Nb. We generate and benchmark printability maps against the experimental data for these commonly printed alloys. We demonstrate that our framework can predict the onset of various defect modes in processing space for arbitrary alloys with acceptable accuracy, precision, and recall. Finally, we demonstrate the framework's effectiveness by predicting printability maps for NiTi-based Shape Memory Alloys (SMAs) at various compositions. The generation of NiTi SMA printability maps is valuable for the fabrication of complex geometries due to their poor machinability[2,7,9,10]. Using the available experimental data for each case study, we also create 'probabilistic printability' maps based on a simple statistical ensembling method described in Section 2.1.1. These case studies highlight our framework's predictive capabilities, enabling accelerated HTP materials and process design in AM.

In the context of AM, printability refers to the processability of an alloy under AM conditions. A printability map shows the different L-PBF processing conditions that lead to various fabrication defects. Lack-of-fusion, balling, and keyholing defect regimes are displayed on the map with colors and bounding lines[3], as depicted in Fig. 1. Typically, a large portion of the process space leads to fabrication defects that compromise the integrity of the final part. Identifying suitable power-velocity combinations for porosity-free parts is crucial for the efficient design of AM experiments. Our proposed framework fulfills this need by identifying potentially defect-free regions in power-velocity space for arbitrary alloys.

Recently, efforts have been made to develop frameworks to assess the printability of arbitrary alloys. The present authors have developed a framework for rapidly evaluating the printability map of arbitrary alloys in L-PBF-AM using physics-based thermal models calibrated with single-track experiments[2]. Islam et al. [11] presented a similar framework, combining analytical models and HTP sample fabrication to determine processing parameters that minimize fabrication defects. However, instead of relying on single-track simulations to construct a printability map, the authors used a dimensionless quantity, $\Pi = \frac{C_p P}{k v^2 h}$, where $C_p$ is the specific heat, $P$ is laser power, $k$ is thermal conductivity, $v$ is laser scan speed, and $h$ is hatch spacing. $\Pi$ captures the heat accumulation in the powder bed and was found to correlate with the final density of the fabricated part. Gordon et al. [12] investigated the impact of processing parameters on porosity formation in L-PBF Ti-6Al-4V using synchrotron-based micro-computed tomography. The study revealed that porosity formation follows predictable trends within the power-velocity space for this material and technique. By examining two Ti-6Al-4V metal AM powders, the study also analyzed gas porosity to determine the transfer of gas pores from the powder to the part during laser melting. A simple Rosenthal model was used to approximate melt pool dimensions and create processing maps to analyze keyholing and lack-of-fusion porosity for fully dense plots. The study investigated one defect at a time and analyzed only one material system.

In addition to physics-based frameworks, machine learning (ML) approaches have been employed to determine alloys' printability. Du et al. demonstrated the use of ML, combined with physics-based models and experimental data, to predict balling in L-PBF-AM[13], creating an empirical indicator for balling. Vela et al. developed a physics-based balling indicator, using it as a feature in ML-models for balling[14]. Both studies focused solely on composition-based balling without addressing other porosity-type defects. Zhu et al.[15] combined dimensional analysis, ML, and experiments to derive a dimensionless quantity related to keyhole porosity formation. Akbari et al.[16] presented MeltpoolNet, a neural network trained with experimental data for predicting metal alloy printability maps. However, the predictions displayed anomalies and insufficient accuracy when compared to experimental data, likely due to sparse training data and a lack of interpretability. Likewise, Scime et al.[6] used feature extraction and unsupervised ML to predict keyholing and balling formation for Inconel 718. Liu et al.[17] used Gaussian Process Regression (GPR) models to predict processing maps and mechanical properties for AlSi10Mg fabricated using L-PBF. The study acknowledges potential limitations, particularly concerning the application of the trained model to materials fabricated in different batches or using different equipment. Variations in equipment settings or powder characteristics may introduce uncertainties in predictions, emphasizing the importance of ensuring consistency in training data sources.

Whalen et al.[18] utilized Bayesian inference to incorporate temperature-dependent material properties and powder bed conditions for a more accurate representation of the L-PBF process. Further refinements were made by calibrating power and speed-dependent effective absorptivity and powder bed porosity values using Bayesian inference, aligning these parameters with experimental data for Inconel 718, Ti-6Al-4V, and 316 Stainless Steel. However, applying Bayesian inference to explore a large alloy space can become computationally expensive, making it impractical for HTP alloy design in L-PBF. The iterative nature of Bayesian methods, particularly when employing Markov Chain Monte Carlo (MCMC) techniques, demands significant computational resources and time, creating a bottleneck for processing large datasets. Additionally, prior information is required for this method, which may not be readily available for new alloys.

In-situ approaches were also considered to predict printability maps. Guirguis et al.[19] introduced an innovative in-situ approach to analyze molten pools in various printing regions and study changes in melt pool shapes for Inconel 718, Ti-6Al-4V, and 316 stainless steel. This approach addresses the critical need for a real-time understanding of metal dynamics during the AM process, where traditional methods often fall short. The method captures temporal features of laser-metal interactions by employing video vision transformers and high-speed imaging, providing detailed insights into the evolving melt pools. These insights enable the construction of precise processing maps that guide the optimization of printing parameters to achieve high-quality results consistently. The results demonstrate significant improvements in defect detection and variability quantification across different alloy compositions and thermofluid properties. While powerful, this framework requires experiments and in-situ melt pool monitoring. While this gives accurate melt pool dimensions, such an approach is inappropriate for HTP screening of printability among candidate alloys.

Although previous works have made significant progress in assessing printability, exploring fully physics-based approaches for predicting printability maps of arbitrary alloys is crucial. Zhu et al.[15] presented a work similar to our contribution. The authors created processing maps for NiTi SMAs using analytical thermal models and defect criteria; however, the approach was not composition-agnostic (i.e., not applied to arbitrary alloys) and cannot be deployed in an HTP manner. Johnson et al.[3] and Zhang et al.[2] demonstrated that printability maps could be derived using finite element methods (FEM) or calibrated Eagar-Tsai models. However, these studies used a single set of defect criteria based on melt pool geometry and processing parameters and had limited experimental data for validation. In contrast, we utilize physics-based models to investigate process maps for various alloys, benchmarking 12 different sets of criteria for the onset of fabrication defects. We compare the generated maps against an extensive validation dataset.

Our work presents a framework with the following aspects: The framework predicts an alloy's printability map without experimental input (disregarding specific thresholds for defect criteria based on melt pool dimensions). The framework predicts the printability map for arbitrary alloys. The framework is parallelized to operate in a HTP manner. The framework considers 12 sets of defect criteria but also provides a weighted ensemble of criteria, which results in a probabilistic printability map. In cases where models agree on a classification, the probability of the defect approaches 1, while in instances of disagreement, the probability of observing the defect tends to be around 0.5.

## Results

### Case study: general AM alloys

To assess the versatility of our framework, we used a diverse dataset of alloys widely investigated in the context of metal AM. Our database comprises both in-house data and data from the literature. Each chemistry-processing point in this database is featurized with approximately 500 descriptors. These descriptors span various aspects, including feedstock conditions, material properties, process parameters, print outcomes, and the performance of the printed parts. Data for 316 Stainless Steel[20–29], Inconel 718[30–38], and Ti-6Al-4V[39–44] were mainly sourced from the published literature. For Ni-5Nb (weight%)[3,8,45] and AF96[4], we combined in-house experiments with data from the literature. The number of data points used to validate the printability maps for each alloy can be found in Table 1. Additionally, the beam diameter, powder layer thickness, and L-PBF machine used for each experimental point are specified.

To evaluate the framework's ability to generate useful printability maps, we produced maps using constant values for layer thickness (30 μm), beam diameter (80 μm), and hatch spacing (70 μm). The true values for these experimental parameters are reported in Table 1. By presenting the true values for these parameters, we transparently cover a large processing space while acknowledging the inherent variability in the data. This approach allows us to see if we can create an effective printability map under standardized conditions, despite having data where these parameters vary.

The ground truth data were constructed by collating the experimental results from both our studies and the literature. Each dataset was carefully vetted to ensure it met our criteria for relevance and accuracy. The data points considered for validation had to meet the following criteria:

- They must be from single-track experiments.
- Both power and velocity points must be provided.
- The state of the as-built fabricated track must be stated or illustrated through microscopic images.
- The power values must be between 40 and 400 W and the velocities between 0.05 and 3.00 m/s.

Based on the features captured in the database such as shown in Fig. 2, the distribution of Linear Energy Density (LED), defined as $LED = P/v$, where $P$ is the laser power and $v$ is the scan speed. This metric is used to print the alloys present in our dataset. To visualize the distribution of a specific feature across different compositions, we employed a combined histogram

**Table 1 | The table displays the number of experimental points used for each alloy to validate the printability maps**

| Alloy Type | Single-Track Datasets | Source | Beam Diameter [μm] | Powder Layer Thickness [μm] | L-PBF Machine |
|---|---|---|---|---|---|
| 316 Stainless Steel | 83 | 20–23,26,27,29 | 52[20], 55[29], 70[23,27] | 40[21], 50[20,23], 75[29], 80[27] | DiMetal-100[21], MCP Realizer SLM250[22], Reinshaw AM400[23] |
| Inconel 718 | 86 | 30,32–34,36–38 | 35[38], 60[16], 70[34], 100[33] | 30[34], 40[30,37], 70[33] | ReaLizer SLM 250[38],BLT-S200[38],EOS M290[33] |
| Ti-6Al-4V | 41 | 39,40,42–44,50 | 70[39,43], 92[42], 100[44,50] | 20[40], 30[40,43,44,50], 40[40], 50[42] | 3D Systems ProX DMP 100[39,43],EOSINT M290[44,50], SLM-500[37] |
| Ni-5Nb | 49 | Experimental data conducted at Texas A&M and published in[3,45] | 100[3,45] | 30[3,45] | 3D Systems ProX DMP 200[3,45] |
| AF96 | 64 | Experimental data conducted at Texas A&M and published in[4] | 80[4] | 37[4] | 3D Systems ProX DMP 200[4] |

The sources, beam diameters, powder layer thicknesses, and L-PBF machines used for each dataset are specified to provide transparency on the experimental conditions for the points used to validate the framework. Some published data did not provide all information about their experimental setup. However, the criteria for data selection was based on whether the data points were for a single-track, power-velocity points were given, and any defect was stated.

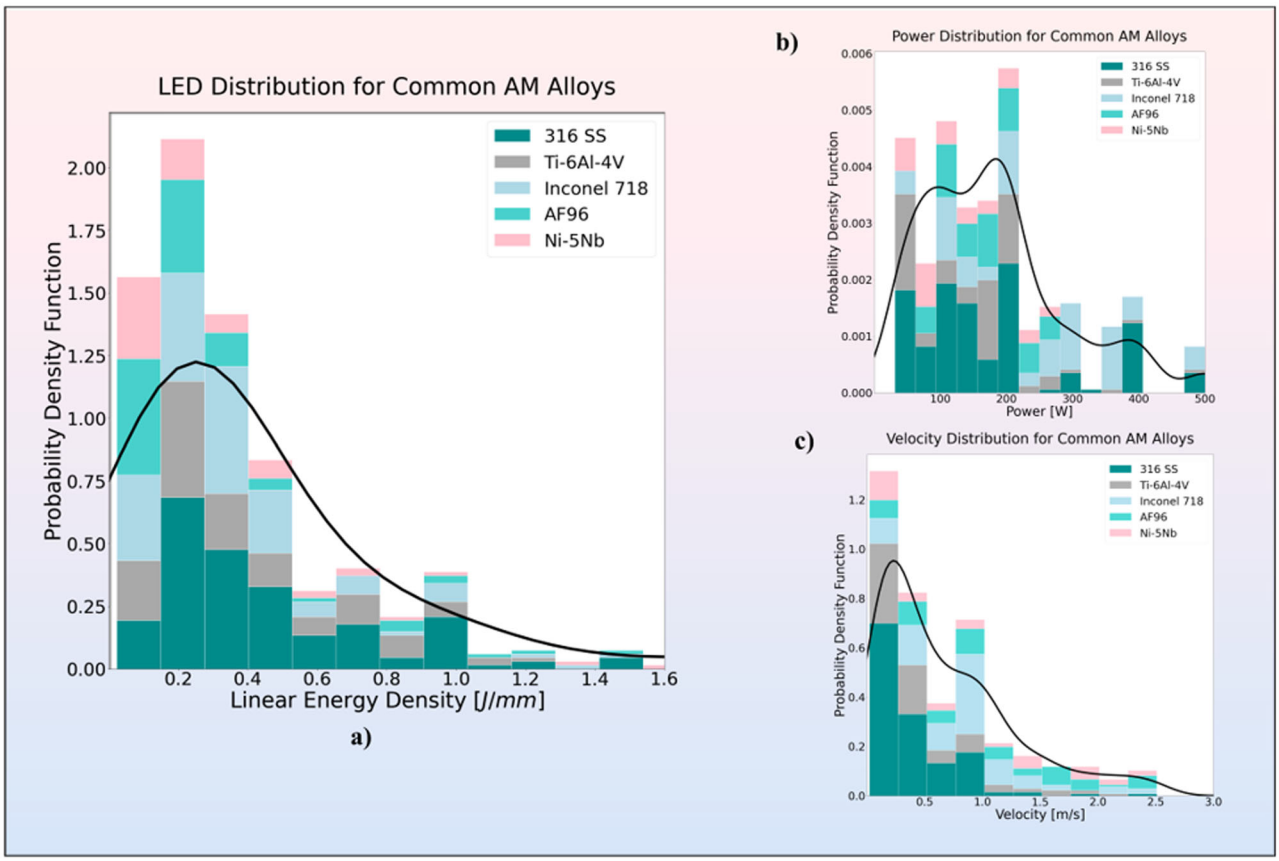



**Fig. 2 | Process parameter distribution for general AM alloys.** The histogram and kernel density estimate (KDE) plots for **a**) Linear Energy Density (LED), **b**) laser power and **c**) scan speed shows the distribution of values used to print for a variety of alloys, as reported in the literature. The density represents the probability density function of the parameter on the x-axis in the plot. AM: Additive manufacturing J: Joules. mm: Millimeter. SS: Stainless Steel.

and kernel density estimation (KDE) approach. The histogram was normalized to form a probability density function (PDF) by dividing the data range into discrete bins, counting the occurrences in each bin, and normalizing by the total number of data points and bin width. This can be mathematically expressed as:

$$f(x) = \frac{1}{n \cdot h} \sum_{i=1}^{n} I\left(x_i \in \text{bin}(x)\right) \quad (1)$$

where $f(x)$ is the estimated density at point $x$, $n$ is the total number of data points, $h$ is the bin width, and $I\left(x_i \in \text{bin}(x)\right)$ is an indicator function that equals 1 if $x_i$ falls into the bin corresponding to $x$ and 0 otherwise.

To further smooth the distribution, we applied kernel density estimation (KDE), which places a Gaussian kernel on each data point and sums the contributions from all kernels. The KDE is given by:

$$\hat{f}(x) = \frac{1}{nh} \sum_{i=1}^{n} K\left(\frac{x - x_i}{h}\right) \quad (2)$$

where $\hat{f}(x)$ is the estimated density at point $x$, $n$ is the total number of data points, $h$ is the bandwidth, and $K$ is the kernel function. For a Gaussian kernel:

$$K(u) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}u^2} \quad (3)$$

The figure shows that the LED peaks between 0.2 and 0.4 $\frac{J}{mm}$ and quickly decays beyond ~ 0.6 $\frac{J}{mm}$. The laser power employed in the materials exhibits a more even distribution, spanning from 50 to ~ 250 W. Scan speed conditions, on the other hand, tend to cluster around relatively low values (< 0.5 m/s), with a significant number of points in the processing space corresponding to high scan speeds between 1 and 2.5 m/s. Based on this analysis, the laser power and scan speed ranges used to generate printability maps for the five alloys mentioned above are 40–400 W and 0.05–3.00 m/s, respectively. The printability maps are generated using 20 equi-spaced power increments and 15 equi-spaced velocity increments.

Regarding other processing parameters not shown on the axes of the printability maps, the diameter of the laser beam, the powder layer thickness, and hatch spacing were held constant at 80 µm, 30 µm, and 70 µm, respectively. These parameters were used to compute the geometry of the melt pool using the thermal model, as described in Section "Calculating the Melt Pool Geometry." Despite being functions of power and velocity, printability maps are affected by other parameters, such as hatch spacing and powder layer thickness. The effects of these processing parameters are discussed in the Supplementary Information and displayed in Supplementary Fig. 1.

In addition to the three defect modes indicated on the printability map, the maximum allowed hatch spacing is also taken into account. The maximum allowed hatch spacing accounts for the required overlap between adjacent tracks to achieve fully dense bulk parts[4]. The maximum allowable hatch spacing is depicted by contour lines on the printability maps.

**Constructing and evaluating printability maps.** In our investigation, we developed printability maps for five different alloys using 12 combinations of defect criteria proposed in the literature, resulting in 60 printability maps in total. All 60 printability maps can be found in the Supplementary Information, where Supplementary Fig. 2 displays the maps for 316 Stainless Steel, Supplementary Fig. 3 display the maps for

Ti-6Al-4V, Supplementary Fig. 4 displays the maps for Inconel 718, Supplementary Fig. 5 displays the maps for AF96 and Supplementary Fig. 6 displays the maps for Ni-5Nb. The performance metrics of these 12 printability maps are summarized in Table 2. In Section "Methods", we describe how these performance metrics are defined. For 316 Stainless Steel, the **LOF1-KH1-Ball1** criteria set displayed a competitive accuracy of 84%, as highlighted in Table 2. Similarly, for Ti-6Al-4V, an acceptable accuracy of 74% was obtained when employing the **LOF2-KH1-Ball1** set. For Inconel 718, the highest accuracy of 78% was achieved with the **LOF2-KH1-Ball2** criteria set. AF96 demonstrated the highest accuracy of 85% when employing the **LOF1-KH3-Ball2** criteria set. When employing **LOF2-KH3-Ball1**, Ni-5Nb displayed an accuracy of 74%. The major difference between those alloys and Ni-5Nb is that for the latter, none of the balling criteria used agreed with experimental observations.

We would also like to note that for most printability maps, the standard deviation of the accuracy ranged from 1% to 4%, except for the maps for Ti-6Al-4V, which had an accuracy standard deviation of 13%. This can be attributed to the fact that some alloys, such as Ti-6Al-4V, contained sparse data for one of the defects (i.e., lack-of-fusion).

It is important to recall that all printability maps were computed assuming fixed laser beam diameters, hatch spacing, and powder bed layer thickness. However, the experimental data correspond to process conditions where these latter parameters varied significantly. The predicted accuracy can thus be considered to be a lower bound set by the fact that we did not use the same process parameters as those reported in experiments. It is expected that when more information on the process conditions is available, the predictive ability of the framework will greatly increase. The comparison with the experimental printability maps for NiTi-based SMAs shown below validates this hypothesis, as all single-tracks on NiTi SMAs were printed under constant processing conditions except for power and velocity.

Based on the analysis above, the **LOF1-KH1-Ball1** criteria set showed effective predictions for 316 Stainless Steel, Ti-6Al-4V, Inconel 718, and AF96. For this reason, we show the printability maps created by the **LOF1-KH1-Ball1** criteria set in Fig. 3. The **LOF1-KH1-Ball1** criteria set has some versatility in predicting the printability of various alloy families. From Fig. 3, it can be seen that the accuracies when using **LOF1-KH1-Ball1** ranged from 84% to 63%. However, Ni-5Nb displays the lowest accuracy due to the fact that the balling criteria did not agree with experimental observations. This discrepancy indicates that the balling indicators used in the current work lack certain physics responsible for the balling phenomena.

The **LOF2-KH1-Ball1** criteria set showed the highest average model probability of 14%. For this reason, we show the printability maps created by the **LOF2-KH1-Ball1** criteria set in Fig. 4. The accuracies for this set of criteria ranged from 83% to 70%. Other criteria sets with similar global accuracies included **LOF1-KH1-Ball2** and **LOF2-KH1-Ball2**. For the four criteria sets with similarly high accuracy, we can see that it is the decision boundary for keyholing remains the same. This indicates that **KH1** can effectively classify keyholing in the processing space.

Dimensional criteria have been verified and utilized throughout the literature as an appropriate method for constructing printability maps. Various studies have demonstrated that dimensional criteria are cost-effective and provide substantial information for optimizing the L-PBF process and creating printability maps.

Balling occurs due to Plateau-Rayleigh capillary instability. Several authors have used the L/W ratio of single-track L-BPF melt pools to quantify this capillary instability. Khairallah et al.[46] used a hybrid FEM/FVM (finite volume method) model to show the role that Length (L)/Width (W), surface tension, and capillary instability have on the balling defect during L-PBF. Bertoli et al.[29] observed that the melt pool elongates and becomes unstable at high laser power and scan speed. Similar behavior was observed in single-track experiments conducted by Li et al.[47]. Yadroitsev et al.[48] showed mathematically how the L/W ratio and the contact angle between the molten track (liquid cylinder) and the substrate where related to the stability of the molten track. These papers all support the use of the L/W ratio as a metric to predict balling.

Lack-of-fusion porosities occur either due to insufficient overlap of melt pools or when the laser power is unable to penetrate the powder bed for complete adhesion. Various studies have explored the effectiveness of using dimensional criteria to predict the quality of AM parts. Tang et al.[49] modeled the cross-section of a melt pool as a half-elliptical shape and determined that full melting occurred when the condition $(\frac{h}{W})^2 + (\frac{t}{D})^2 \leq 1$, where $h$ is the hatch spacing, $W$ is the melt pool width, $t$ is the layer thickness and $D$ is the melt pool depth, is met. Promoppatum et al.[50] verified the usage of this condition to avoid lack of fusion through analytical models and experimental work.

Regarding keyholing, Johnson et al.[3] demonstrated that although keyholing is caused by complex physics involving evaporative effects, it can be diagnosed by small melt pool aspect ratios (W/D). Trapped gases and other porosities induced by keyholing are more likely in melt pools with small W/Depth (D) ratios, as a small W/D ratio indicates excessive superheating of the melt pool. Johnson et al.[3] demonstrated that this metric was useful for classifying keyholing in single-track experiments on AF96 steels.

Regarding printability maps, Seede et al.[4] validated the use of dimensional criteria to construct printability maps for AF9628 parts using L-PBF. The defect regions in the printability maps were defined according to dimensional criteria for lack of fusion, balling, and keyholing. Letenneur et al.[51] used a simple analytical model to predict the density of processed parts for L-PBF. Johnson et al.[3] and Zhang et al.[2] also used dimensional criteria to create and validate their frameworks for process optimization for L-PBF.

By relying on dimensional criteria, we can create practical and effective models that capture essential aspects of the SLM process. These models are cost-effective and align well with observed physical behaviors, providing a solid foundation for predicting and optimizing print quality. Therefore, dimensional criteria serve as a reliable tool for constructing printability maps, enabling accurate and efficient optimization of the SLM process.

There was not a single criteria set that performed the best for all AM alloys, although there were criteria that were slightly more commonly observed to correspond to the onset of certain defects (i.e., **KH1**). Since there is no principled way to establish that a given criteria set is the ground truth, a possible approach is to consider that all the criteria sets considered have a certain degree of validity. The challenge is then to establish a rigorous way to combine the predictions of all the criteria sets simultaneously.

In order to distill the 12 defect criteria into a single printability map, these models were combined using a statistical model averaging approach. The contribution of each model to the ensemble is weighted by its accuracy via Eq. (4). Using this weighting method guarantees the predicted defect probabilities for each processing condition sum to 1. The ensembled printability maps were generated for each of the 5 alloys, as shown in Fig. 5. The opaqueness of each region corresponds to the probability of the said defect occurring during fabrication under a given set of process conditions. When the various criteria sets agree with each other, the probability of the occurrence of a defect increases.

The 12 printability maps will agree with each other in some regions and will disagree with each other in others. In regions where the maps agree, it stands to reason there is a high probability that the prediction is correct whereas in regions where the maps disagree, the prediction would have higher uncertainty. To this end, we fuse information from all 12 printability maps to create a single ensembled probabilistic printability map. However, more accurate printability maps should be given more weight in this final ensembled map. The softmax function can be used to determine these weights. When the accuracies of each of the 12 maps are passed through the softmax function, the softmax function normalizes the accuracies, converting the accuracies into the model probabilities. All model probabilities sum to 1. In this way, the contribution of each map is proportional to the accuracy of the map.

$$\sigma(x)_i = \frac{e^{w*x_{i,c}}}{\sum_{i=1,c=1}^{n} e^{w*x_{i,c}}} \quad (4)$$

**Table 2 | A comparison of the results for the accuracy and model probability across the different general AM alloys that are investigated in this study.**

| Alloy Class: | 316 SS | | Ti64 | | IN 718 | | AF96 | | Ni-5Nb | | Avg. Mod. Prob. |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | Acc. | Mod. Prob. | Acc. | Mod. Prob. | Acc. | Mod. Prob. | Acc. | Mod. Prob. | Acc. | Mod. Prob. | |
| **LOF1-KH1-Ball1** | 84% | 23% | 71% | 23% | 73% | 6% | 81% | 5% | 63% | 1% | **12%** |
| **LOF1-KH1-Ball2** | 82% | 15% | 70% | 15% | 73% | 6% | 85% | 12% | 63% | 1% | **10%** |
| **LOF1-KH2-Ball1** | 83% | 17% | 44% | 0% | 73% | 7% | 83% | 8% | 65% | 2% | **7%** |
| **LOF1-KH2-Ball2** | 79% | 6% | 44% | 0% | 72% | 6% | 83% | 7% | 66% | 3% | **4%** |
| **LOF1-KH3-Ball1** | 80% | 9% | 48% | 0% | 66% | 1% | 83% | 7% | 68% | 5% | **4%** |
| **LOF1-KH3-Ball2** | 80% | 9% | 48% | 0% | 66% | 1% | 85% | 13% | 70% | 7% | **6%** |
| **LOF2-KH1-Ball1** | 76% | 3% | 74% | 41% | 76% | 15% | 83% | 7% | 70% | 7% | **14%** |
| **LOF2-KH1-Ball2** | 75% | 3% | 71% | 21% | 78% | 23% | 82% | 5% | 68% | 5% | **12%** |
| **LOF2-KH2-Ball1** | 81% | 13% | 78% | 78% | 81% | 38% | 80% | 3% | 67% | 4% | **27%** |
| **LOF2-KH2-Ball2** | 81% | 13% | 78% | 78% | 80% | 36% | 82% | 5% | 67% | 4% | **27%** |
| **LOF2-KH3-Ball1** | 81% | 12% | 76% | 68% | 79% | 34% | 83% | 8% | 69% | 5% | **25%** |
| **LOF2-KH3-Ball2** | 82% | 15% | 76% | 69% | 80% | 36% | 84% | 11% | 67% | 4% | **27%** |
| **LOF3-KH1-Ball1** | 82% | 15% | 76% | 69% | 76% | 15% | 84% | 11% | 67% | 4% | **23%** |
| **LOF3-KH1-Ball2** | 82% | 15% | 76% | 69% | 76% | 15% | 84% | 11% | 67% | 4% | **23%** |
| **LOF3-KH2-Ball1** | 84% | 24% | 76% | 69% | 80% | 36% | 84% | 11% | 68% | 5% | **29%** |
| **LOF3-KH2-Ball2** | 84% | 24% | 76% | 69% | 81% | 38% | 84% | 11% | 67% | 4% | **29%** |
| **LOF3-KH3-Ball1** | 82% | 15% | 76% | 69% | 80% | 36% | 83% | 8% | 66% | 3% | **26%** |
| **LOF3-KH3-Ball2** | 82% | 15% | 76% | 69% | 80% | 36% | 84% | 11% | 66% | 3% | **26%** |

The model probability is calculated using Eqn. (4). The average model probability associated with **LOF1-KH1-Ball1/Ball2** and **LOF2-KH1-Ball1/Ball2** are 10% or above, allowing us to conclude that these criteria sets can represent the printability space for the general AM alloys. These criteria sets help with an initial prediction of where in the processing space a defect-free product can be fabricated using L-PBF AM.

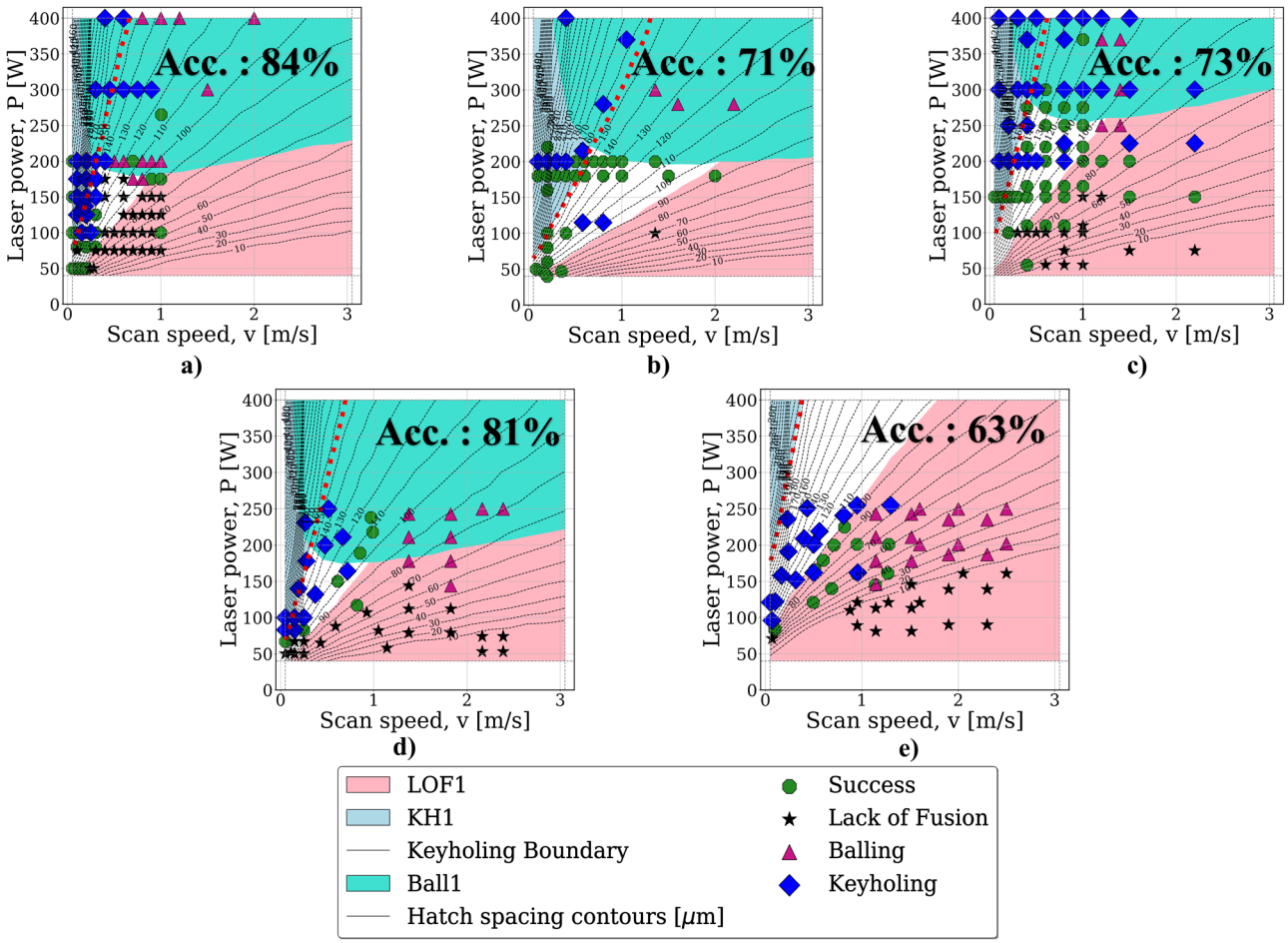



**Fig. 3 | Printability map with LOF1-KH1-Ball1 criteria set for the general AM alloys.** The printability maps shown are for the following alloys: **a**) 316 Stainless Steel, **b**) Ti-6Al-4V, **c**) Inconel 718, **d**) AF96 and **e**) Ni-5Nb. In general, **LOF1-KH1-Ball1** criteria set, defined in Table 6 showed to be the most versatile combination of the criteria set. The accuracies ranged from 63% to 84%, where Ni-5Nb showed the lowest value for accuracy. LOF Lack-of-fusion, KH Keyholing, Ball: Balling, Salmon: Regions predicted for lack-of-fusion to occur. Blue: Region in processing space for keyholing to occur. Teal: Region predicted for balling to occur. Green circles: Experimentally observed defect-free single tracks. Black star: Experimentally observed lack-of-fusion. Magenta triangles: Experimentally observed balling. Blue diamond: Experimentally observed keyholing. Red dashed line: Keyholing boundary. Black dashed lines: Maximum hatch spacing required for sufficient overlap in a track in fully built parts.

In Eq. (4), $x_{i,c}$ is the accuracy calculated for each defect for each criteria set and $w$ is a weight factor. The weight factor determines how probable the criteria set is and can be adjusted based on the experimenter's analysis and a priori knowledge of the criteria set. However, to bias one criteria set over another, the weight factor, $w$, would need to be exceptionally high. Therefore, a considerable amount of data that verifies the boundaries for the defect regions would be needed to bias one criteria set over any of the others. By predicting the model probability associated with each criteria set, insight into the uncertainty associated with each criteria set and printability map is taken into account when deciding the optimal criteria set for a given alloy. Using 316 Stainless Steel as an example problem, the calculation for the uncertainty associated with each criteria set for the lack-of-fusion can be defined as shown in Table 3.

By plotting each of the criteria combination sets with the uncertainty values, Fig. 5 can be constructed.

In Fig. 5a, it is evident that the printable window for 316 SS is very narrow. However, the probability of keyholing occurring decreases closer to the center of the map, near the printable region. The same trend can be observed for balling. Furthermore, the maps for Inconel 718 in Fig. 5c and AF96 in Fig. 5d also show a narrow printable region. The area predicted for lack-of-fusion and keyholing is much larger but less prominent in areas near the center of the maps for both alloys as well. In contrast, the map for Ti-6Al-4V in Fig. 5b shows a larger printable region in comparison to the other alloys. Lastly, in the probabilistic printability map for Ni-5Nb in Fig. 5e, the balling criteria used in the evaluation for the defect did not match the experimental observation. Again, this indicates that the current balling indicator is missing physics to explain the phenomena in Ni-5Nb.

**Analysis of balling criterion**. In the analysis of the balling region in Ni-5Nb, a detailed investigation into the sensitivity of the criteria was undertaken. The criterion, Ball2, can be expressed as L/W, where L is the length of the melt pool and W is the width of the melt pool. The ratio has a threshold value of 3.85, and for ease of analysis, only Ball1 was considered. Figure 3e, depicting Ball1, did not reveal a visible region indicating balling. However, a significant development occurred in Fig. 6 when the power increased to 700 W from 400 W, and the scan speed rose to 5.0 m/s from 3.0 m/s, resulting in the emergence of a visible balling region on the printability map.

The balling criteria were defined based on the ratio of melt pool length (L) to melt pool width (W), where a high ratio indicated the occurrence of Plateau-Rayleigh instability, increasing the likelihood of the melt track breaking into droplets. Conducting a sensitivity analysis for the threshold values of Ni-5Nb revealed that as the threshold value increased, the material's susceptibility to balling decreased. The more restrictive threshold conditions demanded a larger difference between L and W to signify a balling region. This stringency implied that the material was less likely to

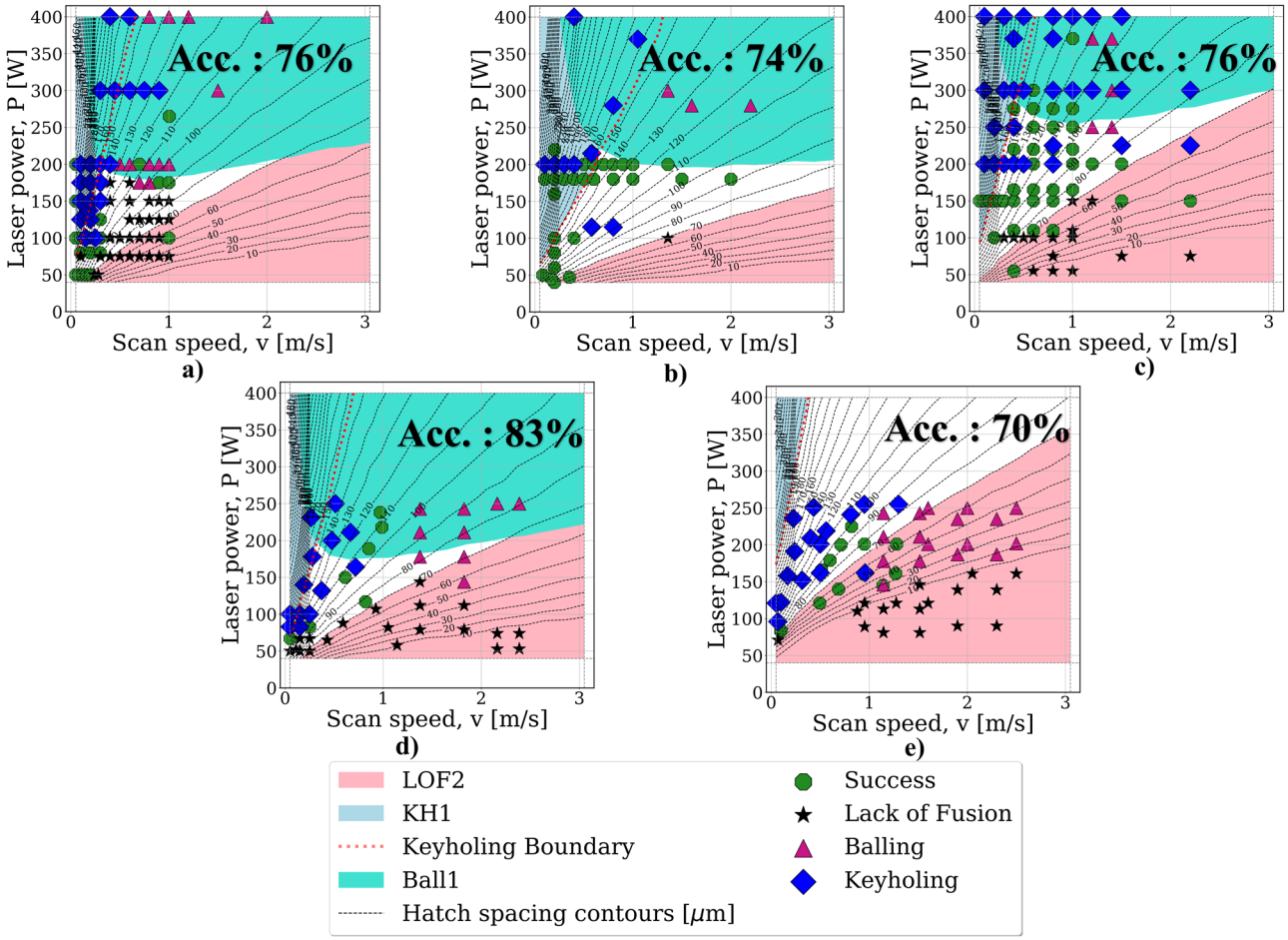



**Fig. 4 | Printability map with LOF2-KH1-ball1 criteria set for the general AM alloys.** The printability maps shown are for the following alloys: **a**) 316 Stainless Steel, **b**) Ti-6Al-4V, **c**) Inconel 718, **d**) AF96, and **e**) Ni-5Nb. Based on the accuracies for each AM alloy, as well as the average model probability for **LOF2-KH1-Ball1**, the criteria set can be considered as one of the criteria sets that gives a good prediction of the printability space for the alloys. The accuracies ranged from 70% to 83%, with Ni-5Nb having the lowest accuracy value. LOF: Lack-of-fusion. KH: Keyholing. Ball: Balling. Salmon: Regions predicted for lack-of-fusion to occur. Blue: Region in processing space for keyholing to occur. Teal: Region predicted for balling to occur. Green circles: Experimentally observed defect-free single tracks. Black star: Experimentally observed lack-of-fusion. Magenta triangles: Experimentally observed balling. Blue diamond: Experimentally observed keyholing. Red dashed line: Keyholing boundary. Black dashed lines: Maximum hatch spacing required for sufficient overlap in a track in fully built parts.

classify an area in the printability map as a region prone to balling. Therefore, a less stringent threshold value was necessary for Ni-5Nb to identify an area for balling for the processing parameter ranges in Fig. 3.

Initially, a threshold value of 2.3 was applied consistently for the printability analysis of various materials. However, after a comprehensive analysis of the material properties of Ni-5Nb and four other materials, it became evident that Ni-5Nb exhibited slightly more resistance to balling due to the interplay of various material properties. Thermal properties, including thermal conductivity, density, and surface tension, were analyzed. The thermal conductivity values at the liquidus temperature ranged from 15.8 W/mK to 29.9 W/mK, with Ni-5Nb having the highest and 316 Stainless Steel the lowest. Higher thermal conductivity contributes to a more uniform energy distribution within the molten pool, reducing the chances of instabilities and balling.

Density calculations showed Ni-5Nb having the highest density at 8.89 g/cm$^3$, contributing to higher energy absorption and reduced balling susceptibility. The effective specific heat capacity, calculated as 742.0 J/kgK for Ni-5Nb and 983.8 J/kgK for 316 SS, indicated a lower effective specific heat capacity for Ni-5Nb, allowing for efficient laser energy absorption, complete melting of the powder bed, and controlled melt pool formation. Additionally, Ni-5Nb exhibited higher surface tension at 1.78 N/m compared to 316 SS at 1.37 N/m, promoting a stable molten pool dynamic and increasing resistance to balling.

In conclusion, the absence of a visible balling region in Fig. 3e for Ni-5Nb, compared to other materials, is attributed to the intricate interplay between material properties and processing conditions. The analysis underscores the importance of understanding these interactions for optimizing additive manufacturing processes and minimizing balling defects. The analysis also highlights the sensitivity of the balling criteria to material properties and processing conditions, indicating a need for a more universal balling criterion.

### Case study with NiTi-alloy system

To further show the effectiveness of the framework, we selected three compositions from the NiTi alloy family, and their printability maps were generated and evaluated. The three NiTi alloys chosen were: $Ni_{50.1}Ti_{49.9}$ (at. %), $Ni_{50.8}Ti_{49.2}$ (at. %), and $Ni_{51.2}Ti_{48.8}$ (at. %). The selected alloys were chosen because our groups extensively studied their printability in previous work[7,52], with each alloy undergoing 60 single-track experiments. These experiments are illustrated in Fig. 7. Contrary to the benchmarking of the predicted printability maps of the 5 alloys above against literature data, in this case, we had full knowledge of the process conditions, including laser beam diameter and powder layer thickness, fixed at 30 μm and 80 μm, respectively. This information serves as a definitive ground truth to evaluate our framework, as the uncertainty arising from unknown or unaccounted-for process conditions is minimized.

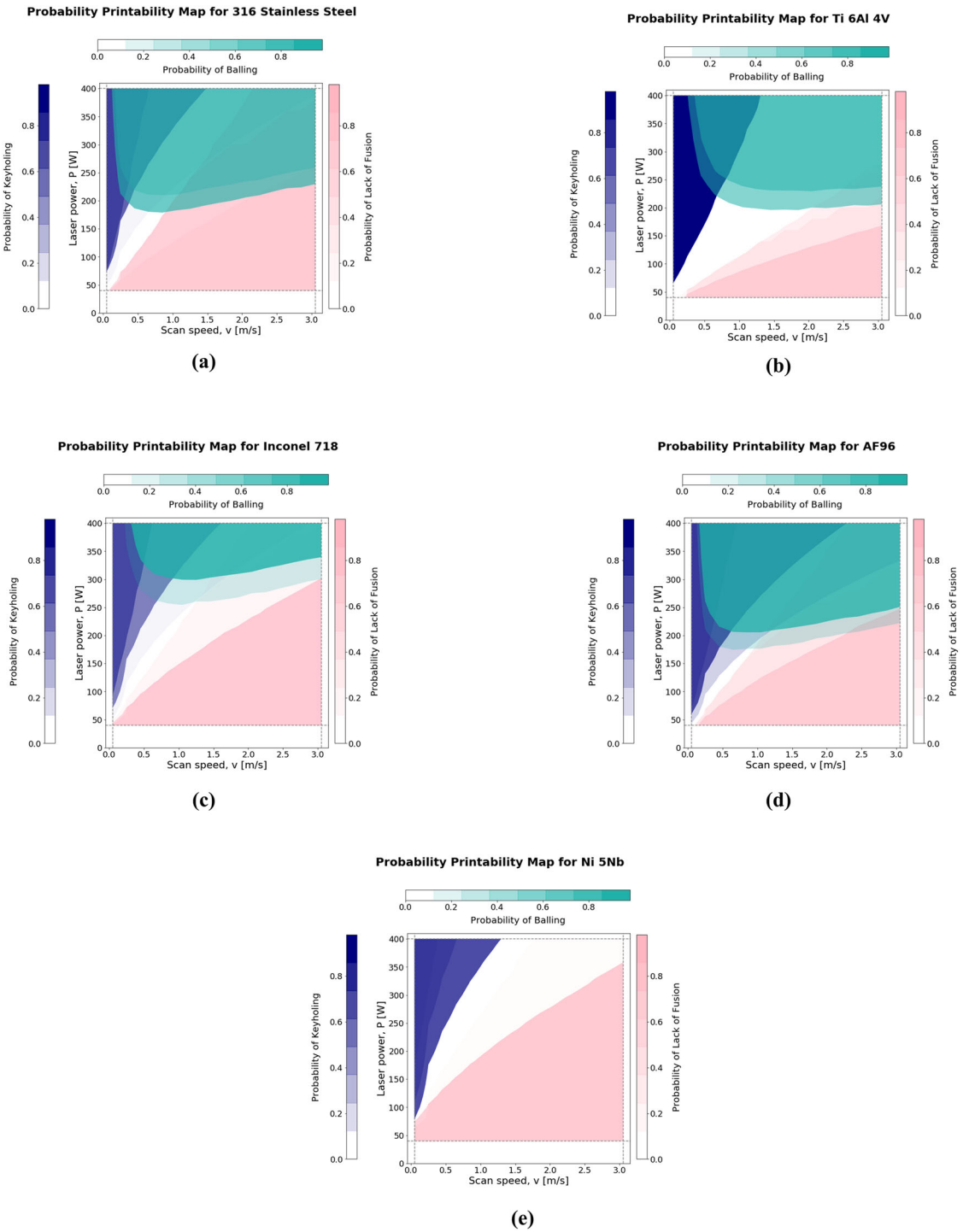



**Fig. 5 | Probability printability maps for general AM alloys.** The ensemble probabilistic printability maps for the following allows are displayed as the following: **a**) 316 Stainless Steel, **b**) Ti-6Al-4V, **c**) Inconel 718, **d**) AF96 and **e**) Ni-5Nb. Using the accuracies of the 12 different criteria sets, the probability of the criteria set was calculated with Eqn. 4. The opaqueness of each region corresponds to the probability of said defect occurring at a particular laser power and speed. LOF: Lack-of-fusion. KH: Keyholing. Ball: Balling. Salmon: Regions predicted for lack-of-fusion to occur. Blue: Region in processing space for keyholing to occur. Teal: Region predicted for balling to occur. Green circles: Experimentally observed defect-free single tracks. Black star: Experimentally observed lack-of-fusion. Magenta triangles: Experimentally observed balling. Blue diamond: Experimentally observed keyholing. Red dashed line: Keyholing boundary. Black dashed lines: Maximum hatch spacing required for sufficient overlap in a track in fully built parts.

**Determining process parameters**. To define the processing design space for NiTi alloys, the range of processing parameters for single-track experiments on NiTi available in the literature were analyzed. Figure 8 displays the range of laser powers and scan speeds used to print NiTi alloys published in the literature. Based on the available data, for the case study of NiTi-based alloy, the processing range considered was 40–300 W and 0.08–2.33 m/s for laser power and scan speed, respectively.

Figure 8 also shows that the distribution of volumetric energy densities (VEDs) defined as $VED = \frac{P}{v \cdot h \cdot t}$, peaks within the 50–100 $\frac{J}{mm^3}$ range and quickly decays in frequency as soon as VED exceeds $\sim$ 150 $\frac{J}{mm^3}$. In contrast,

**Table 3 | The LOF accuracy per criteria combination set and the calculated $\sigma(x)_c$ for 316 Stainless Steel.**

| Criteria Combination | LOF Accuracy | $\sigma(x)_c$(%) |
|---|---|---|
| LOF1-KH1-Ball1 | 0.8674 | 15 |
| LOF1-KH1-Ball2 | 0.8434 | 9 |
| LOF1-KH2-Ball1 | 0.8554 | 12 |
| LOF1-KH2-Ball2 | 0.8675 | 15 |
| LOF1-KH3-Ball1 | 0.8554 | 12 |
| LOF1-KH3-Ball2 | 0.8795 | 19 |
| LOF2-KH1-Ball1 | 0.7711 | 2 |
| LOF2-KH1-Ball2 | 0.7831 | 3 |
| LOF2-KH2-Ball1 | 0.7831 | 3 |
| LOF2-KH2-Ball2 | 0.7831 | 3 |
| LOF2-KH3-Ball1 | 0.7831 | 3 |
| LOF2-KH3-Ball2 | 0.7831 | 3 |

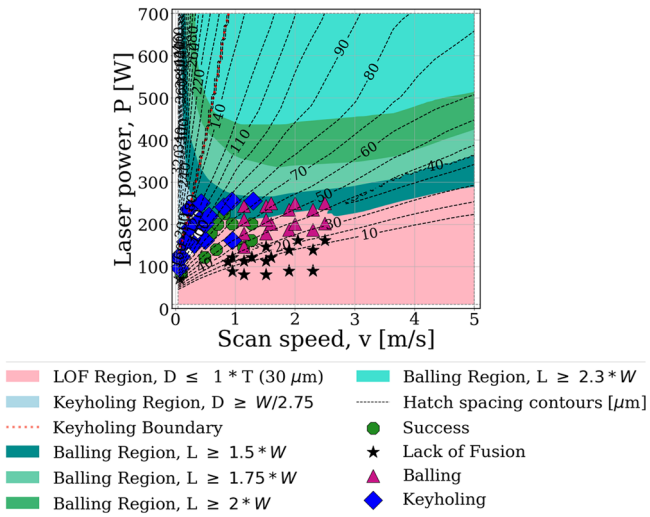



**Fig. 6 | Ni-5Nb balling sensitivity analysis.** A sensitivity analysis was performed on the balling criteria for Ni-5Nb, revealing insightful findings. As the threshold value for L/W increased, the criteria became more stringent, resulting in a reduction of the area identified as exhibiting balling. Interestingly, a broader processing parameter range was employed, demonstrating a balling region for a threshold value of 2.3. This observation implies that Ni-5Nb required a less stringent condition for balling, highlighting its lower susceptibility compared to other materials. T: Powder layer thickness. D: Melt pool depth. W: Melt pool width. L: Melt pool length. Green circles: Experimentally observed defect-free single tracks. Forest Green: Boundary constraint for balling at 1.5. Mint Green: Boundary constraint for balling at 1.75. Green: Boundary constraint for balling at 2. Teal: Boundary constraints for balling at 2.3. Black star: Experimentally observed lack-of-fusion. Magenta triangles: Experimentally observed balling. Blue diamond: Experimentally observed keyholing. Red dashed line: Keyholing boundary. Black dashed lines: Maximum hatch spacing required for sufficient overlap in a fully built part.

the distribution of laser powers is more evenly distributed, ranging from 50 to ~ 250 W. The distribution of scan speeds is skewed toward relatively low values (< 0.5 m/s), although there are some instances of high scan speeds between 1–2 m/s. It should be noted that the exploration of the processing space has been exhaustive for some alloys. However, in other cases, such as $Ni_{50.9}Ti_{49.1}$, the alloys were fabricated within a very narrow process window.

**Constructing and evaluating printability maps**. For the three selected NiTi alloys, a 10 × 6 grid of experiments for laser power and scan speed

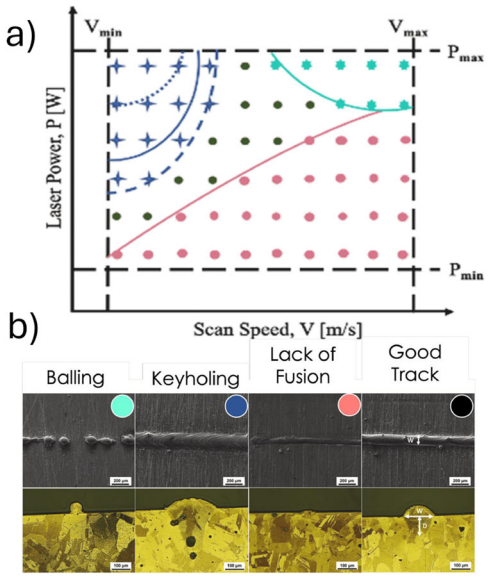



**Fig. 7 | Schematic of the ground truth process space for NiTi alloy experiments. a)** Schematic, in process space, of `ground truth' experiments for the dataset developed in-house. Experiments were designed on a grid containing 60 points per alloy. **b)** Morphology and cross-sections for each of the single-track experiments were used to classify different points in the process space in terms of the type of defect. Schematic and micrographs were adapted from the original reference [2]. vmin: Minimum scan speed. vmax: Maximum scan speed. Pmin : Minimum power. Pmax : Maximum power. Salmon circles: Predicted processing points for lack-of-fusion. Blue diamonds: Predicted processing points for keyholing. Teal stars: Predicted processing points for balling. Black circles: Predicted points for defect-free.

was used to query the melt-pool dimensions for the printability maps. The maps for each combination of the criteria were plotted for all NiTi alloys and can be found in the Supplementary Information, where Supplementary Fig. 7 displays the printability maps for $Ni_{50.3}Ti_{49.7}$, Supplementary Fig. 8 displays the printability maps for $Ni_{50.8}Ti_{49.2}$,Supplementary Fig. 9 displays the printability maps for $Ni_{51.2}Ti_{48.8}$.

For $Ni_{50.3}Ti_{49.7}$, the accuracy of the maps ranged from 82% to 84%, as shown in Table 4. For $Ni_{50.8}Ti_{49.2}$, the accuracy of the maps ranged from 81% to 87%, while for $Ni_{51.2}Ti_{48.8}$, the accuracy ranged from 78% to 91%. The variance associated with the accuracy values over the 12 printability maps was 5% or less. The low variance can be attributed to the fact that the machine parameters were consistent for each of the alloys. The most accurate criteria sets for the 3 NiTi alloys **LOF2-KH1-Ball1** and **LOF2-KH2-Ball1**.

The criteria set with the highest average model probability (15%) is **LOF2-KH2-Ball1**. As seen in Fig. 9, using this criteria set, the accuracies for the three alloys were 84%, 87%, and 91% for $Ni_{50.3}Ti_{49.7}$, $Ni_{50.8}Ti_{49.2}$, and $Ni_{51.2}Ti_{48.8}$, respectively.

Once the error metrics associated with each criteria set were determined, these 12 defect criteria were assembled into a single probabilistic model, as we did above for the 5 benchmarking alloys. The contribution of each criteria set to the ensemble is weighted by its accuracy as described in Section "Constructing and Evaluating Printability Maps." These ensembled printability maps show the probability of defects occurring throughout the power-velocity space. The printable region (0% probability of defects) is very narrow for these 3 alloys, as shown in Fig. 10. The probability of the defects occurring decreases toward the center of the map. This is because the models have a consensus that the region is defect-free. Likewise, the probability of defects occurring increases when the models agree that the region contains a certain defect, i.e., in the bottom right corner, all models agree that these parameters result in lack-of-fusion.

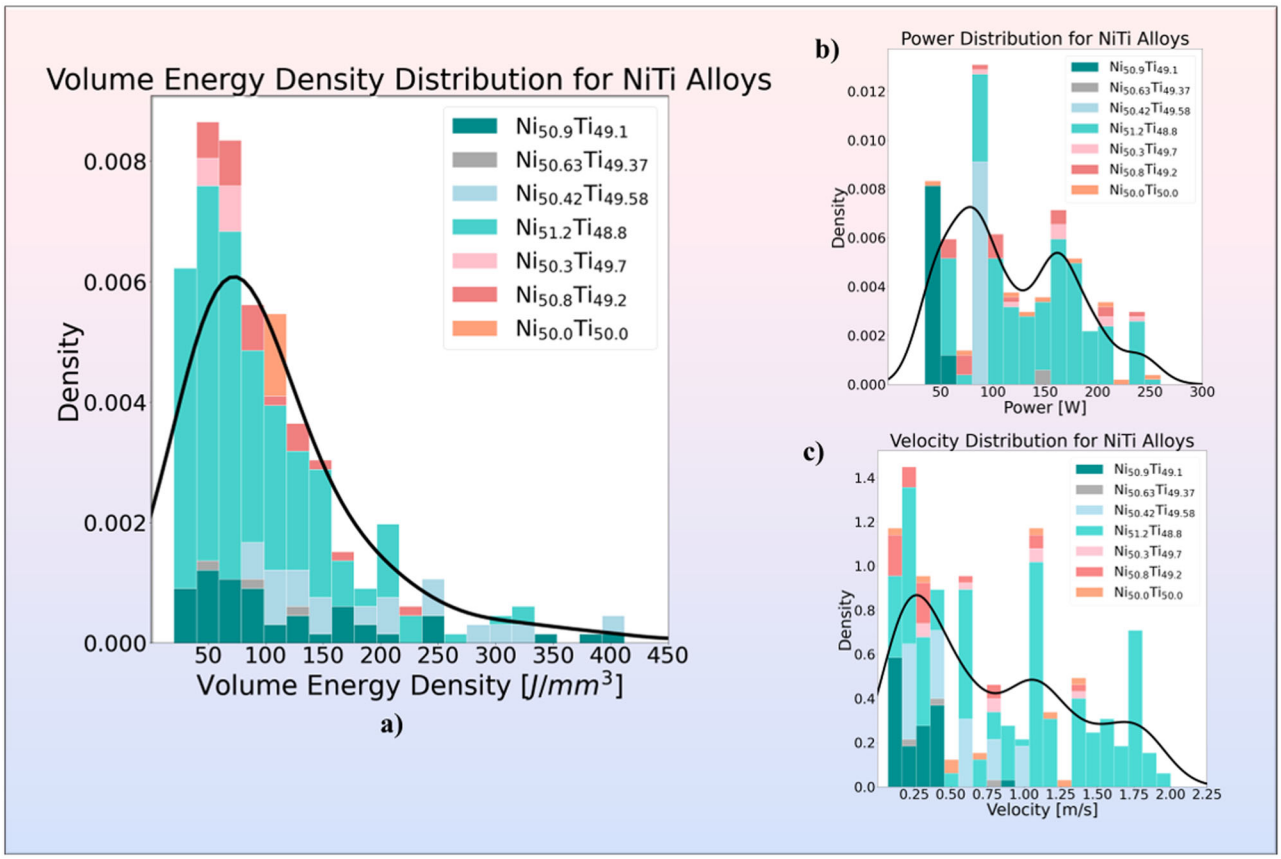



**Fig. 8 | Process Parameter Distribution for NiTi Alloys.** The histogram and kernel density estimate (KDE) plots for **a**) Volume Energy Density (VED), **b**) laser power, and **c**) scan speed shows the distribution of values used to print for various NiTi-based alloys, as reported in the literature. The density represents the probability density function of the parameter on the x-axis in the plot.

**Table 4 | A comparison of the results for the accuracy and model probability across the NiTi alloys**

| Composition: | $Ni_{50.3}Ti_{49.7}$ | | $Ni_{50.8}Ti_{49.2}$ | | $Ni_{51.2}Ti_{48.8}$ | | Avg. Mod. Prob. |
|---|---|---|---|---|---|---|---|
| | Acc. | Mod. Prob. | Acc. | Mod. Prob. | Acc. | Mod. Prob. | |
| **LOF1-KH1-Ball1** | 84% | 9% | 84% | 7% | 80% | 2% | **6%** |
| **LOF1-KH1-Ball2** | 82% | 6% | 82% | 4% | 78% | 1% | **4%** |
| **LOF1-KH2-Ball1** | 84% | 9% | 84% | 7% | 80% | 2% | **6%** |
| **LOF1-KH2-Ball2** | 82% | 6% | 82% | 4% | 81% | 3% | **4%** |
| **LOF1-KH3-Ball1** | 84% | 9% | 84% | 7% | 80% | 2% | **6%** |
| **LOF1-KH3-Ball2** | 82% | 7% | 81% | 4% | 79% | 2% | **4%** |
| **LOF2-KH1-Ball1** | 84% | 10% | 87% | 13% | 90% | 17% | **13%** |
| **LOF2-KH1-Ball2** | 83% | 8% | 86% | 11% | 90% | 16% | **12%** |
| **LOF2-KH2-Ball1** | 84% | 10% | 87% | 14% | 91% | 21% | **15%** |
| **LOF2-KH2-Ball2** | 84% | 9% | 86% | 11% | 90% | 16% | **12%** |
| **LOF2-KH3-Ball1** | 84% | 10% | 85% | 8% | 87% | 9% | **9%** |
| **LOF2-KH3-Ball2** | 84% | 8% | 84% | 7% | 86% | 8% | **8%** |
| **Min/Avg/Max/Std Dev Acc.** | **82 — 83 — 84 — 1%** | | **81 — 84 — 87 — 2%** | | **78 — 84 — 91 — 5%** | | |

The model probability is calculated using Equation (4). The accuracy is calculated using Equation (18).The average model probability for **LOF2-KH1-Ball1** and **LOF2-KH2-Ball1** are 13% and 15%, respectfully, allowing us to conclude that these criteria set combinations are universal for the NiTi -based alloys.

It is evident that the ensemble model exhibits uncertainty regarding the boundaries of the balling region, as indicated by the diminished opacity in the green areas. This is because the balling criteria had the worst classification metrics of any class. In the different printability maps displayed in Fig. 9, most points labeled as belonging to the balling region are actually at the boundary between balling and the other regions of the processing map (keyholing, lack-of-fusion, printable regions). Thus, prediction for this specific region was very sensitive to small inaccuracies.

Details on the specific performance metrics can be visualized with confusion matrices. The confusion matrices for the best-performing criteria

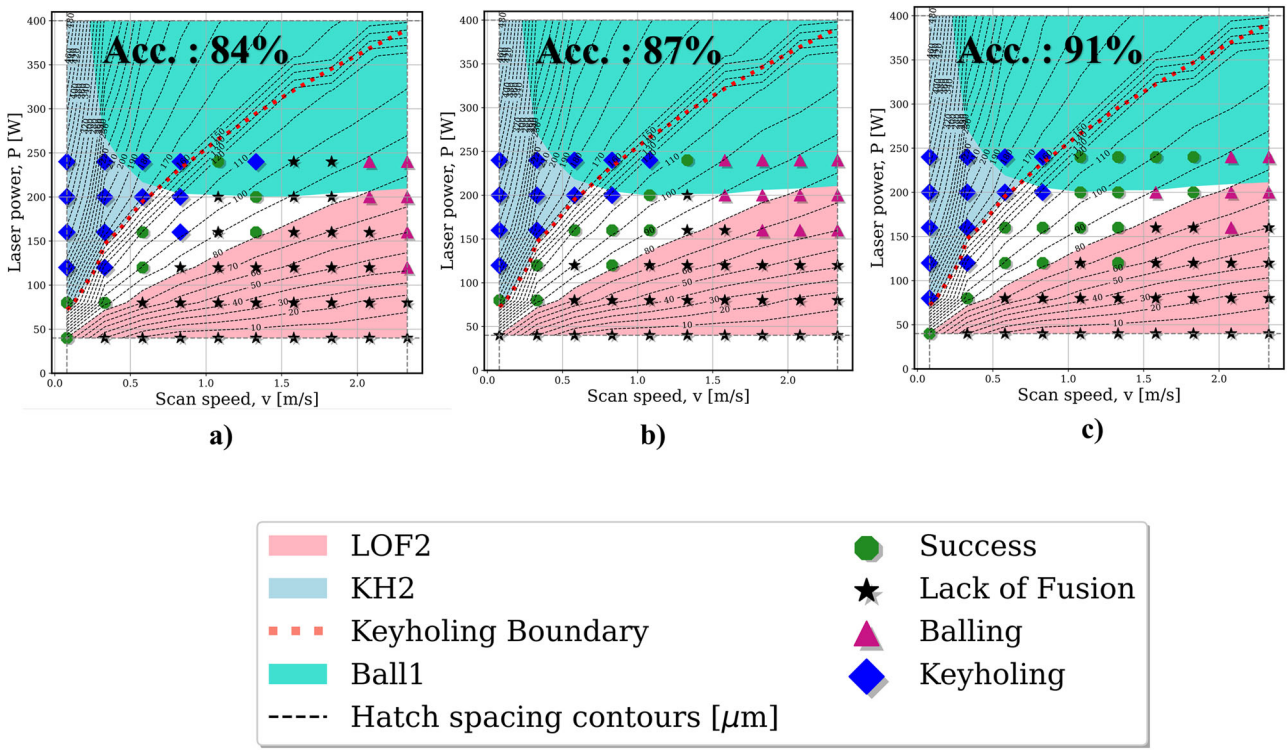



**Fig. 9 | Printability map with highest accuracy for NiTi alloys.** Printability maps for **a**) $Ni_{50.3}Ti_{49.7}$ **b**) $Ni_{50.8}Ti_{49.2}$and **c**) $Ni_{51.2}Ti_{48.8}$ were developed. For the criteria set, **LOF2-KH2-Ball1**, the accuracies for the NiTi compositions ranged from 84% to 91%. The criteria set also had the highest average model probability at 15%, indicating that it was one of the criteria sets that was able to predict the boundary lines for the porosity-induced defects more closely than the other criteria sets. Salmon: Regions predicted for lack-of-fusion to occur. Blue: Region in processing space for keyholing to occur. Teal: Region predicted for balling to occur. Green circles: Experimentally observed defect-free single tracks. Black star: Experimentally observed lack-of-fusion. Magenta triangles: Experimentally observed balling. Blue diamond: Experimentally observed keyholing. Red dashed line: Keyholing boundary. Black dashed lines: Maximum hatch spacing required for sufficient overlap in a track in fully built parts.

set (**LOF2-KH2-Ball1**) are shown in Fig. 11 as an example. The following classification results are shown in the confusion matrices:

- True positive (top-left corner of matrix)
- False negative (top-right corner of matrix)
- False positive (bottom-left corner of matrix)
- True negative (bottom-right corner of matrix)

As detailed in Section "Evaluating the predicted printability maps," these values can be used to calculate precision, recall, and accuracy for each defect, based on Equations (16), (17), and (18). In the case of $Ni_{50.3}Ti_{49.7}$ alloy, the accuracy for lack-of-fusion, keyholing, balling, and the printable region were 81%, 92%, 83%, and 83%. The accuracy for the binary classifications of defects for the $Ni_{50.8}Ti_{49.2}$ alloy was 87%, 95%, 82%, and 87% for lack-of-fusion, keyholing, balling, and the printable region. For the $Ni_{51.2}Ti_{48.8}$ alloy, the accuracy for lack-of-fusion, keyholing, balling, and the printable region were 93%, 97%, 83%, and 87%.

While the proposed framework is not perfectly predictive, the resulting printability maps in Section "Case study: general AM alloys" and Section "Case study with NiTi-alloy system" tend to offer relatively good agreement with experiments and constitute a valuable first step towards the definition of more precise printability regions. Moreover, this framework does not require experimental input. The framework only requires an alloy composition and a range of processing parameters and can predict the printability of an alloy a priori, provided one has reliable models for the relevant thermo-physical properties.

Furthermore, the proposed framework provides interpretable and safe-to-extrapolate predictions that are free of the anomalies that other approaches suffer from[16], such as discontinuities and other irregularities in the processing maps. Furthermore, we account for 12 different combinations of defect criteria and assemble them into a probabilistic printability map.

## Discussion

In order to better assess the relative value of the proposed predictive framework, it is interesting to compare the resulting printability maps to what would be obtained through a more laborious calibrated printability map. Generally, to construct a calibrated printability map[15], an analytical model, such as the E-T model, is implemented and calibrated against experimental data on the melt pool geometry at different process conditions. Then, on the basis of the melt pool geometry from the E-T analytical model, the boundaries and regions in the processing space are determined using the experimentally derived criteria for three manufacturing defects—i.e., lack-of-fusion, balling, and keyholing.

For example, to construct the calibrated printability map for $Ni_{50.3}Ti_{49.7}$, shown in Fig. 7a, approximately 10 days of work are required due to the need to perform single-track experiments, measure the dimensions of the resulting melting pool and then calibrate the E-T model. On the contrary, the generation of the corresponding computationally based printability map in Fig. 7b just takes around 2-3 hours by the framework described in Section "Case study with NiTi-alloy system"—we note that this computational cost can potentially be reduced to minutes when using ML-based analogs to the E-T model, as will be shown in future work. Moreover, as observed in Fig. 12, the classification performance metrics calculated for these two maps are very similar. Therefore, the results of our computational framework are almost as accurate as those of the previously generated calibrated printability maps while being significantly more resource-efficient.

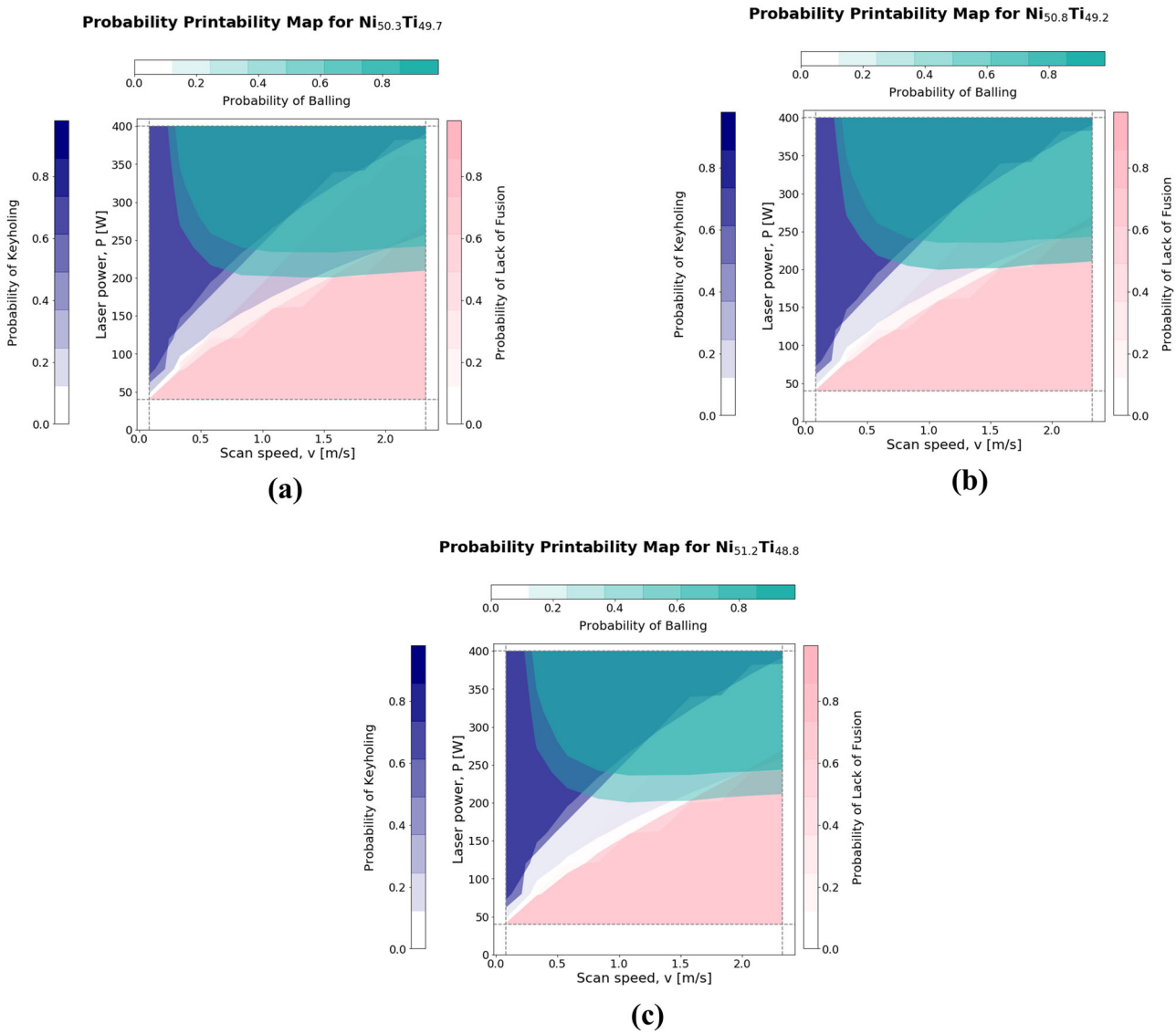



**Fig. 10 | Probability printability maps for NiTi alloys.** Probabilistic printability maps were created for **a**) $Ni_{50.3}Ti_{49.7}$ **b**) $Ni_{50.8}Ti_{49.2}$and **c**) $Ni_{51.2}Ti_{48.8}$. Based on the three binary classification problems based on defect presence (i.e., lack-of-fusion, keyholing, and balling) for each defect criterion, the probability of the region of the defect occurring is defined by how opaque the color of the region is. From all three maps, we can see that the criterion for keyholing covers a wide area, reducing the printable region. Salmon: Regions predicted for lack-of-fusion to occur. Blue: Region in processing space for keyholing to occur. Teal: Region predicted for balling to occur.

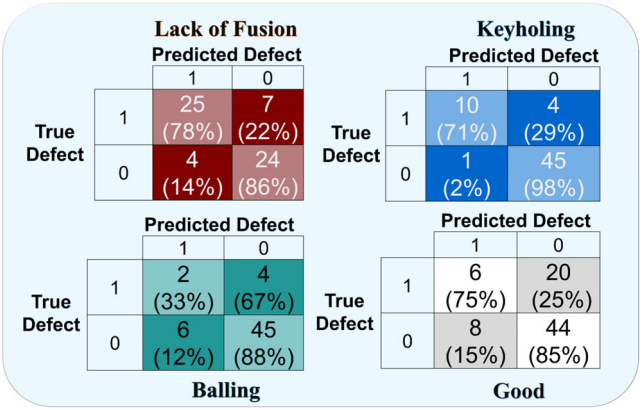
This provides confidence in the use of this framework for HTP analysis of the printability of the entire alloy spaces. We envision future use cases of the proposed computational framework where hundreds or even thousands of alloys can be evaluated for their susceptibility to the onset of AM fabrication defects ahead of their experimental characterization. Such a framework would constitute an important tool within any ICME-based approach to the design of alloys for performance *and* printability. Importantly, and in contrast with data-only approaches, the boundaries within the printability map can be easily interpreted and directly connected to specific combinations of process conditions and material properties. This is particularly true when considering ML frameworks based on complex models such as neural networks, as they tend to have poor extrapolation performance over unseen regions of the processing space.

## Methods

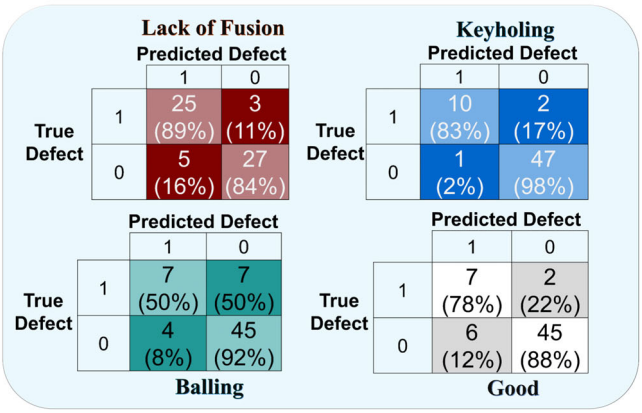
The proposed framework consists of four crucial stages: thermo-physical property calculation, melt pool dimension calculation, defect criteria selection, and printability map construction, as illustrated in Fig. 13. Regarding inputs, the framework requires alloy chemistry and processing conditions (i.e., laser power and scan speed ranges considered in the design space). The steps are as follows: 1) Thermo-physical properties for the alloys in question are queried from CALPHAD models and fed to the thermal model. 2) The thermal model simulates single-track prints for all power-velocity pairs and returns the resultant melt pool dimensions. 3) The melt pool dimensions are then compared to criteria to classify the print as defect-free or not. 4) Finally, these classifications are mapped back onto the power-velocity space, creating the printability map. Each stage of the framework is discussed in further detail below.

### Calculation of thermophysical properties and melt pool geometry

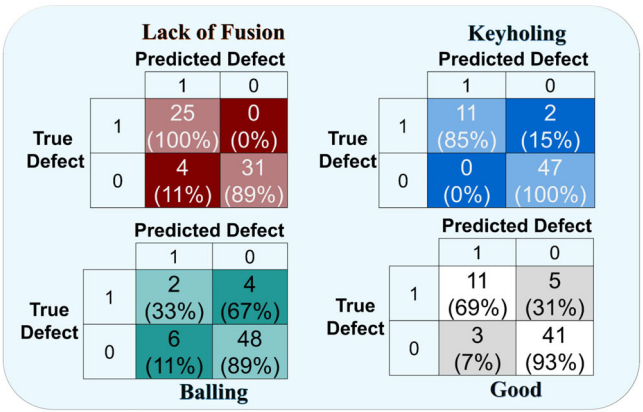
To calculate the thermo-physical properties relevant to AM, several models and formulations were used. Our framework uses Thermo-Calc's Property and Equilibrium modules to estimate thermo-physical properties. While multiple CALPHAD databases exist for specific alloy systems (e.g., TCHEA5, TCAL7, TCFE10, and TCNI11), for simplicity and consistency, we opted to use the most generalized CALPHAD database, Thermo-Calc's high entropy alloy database TCHEA5. Details on how to query the thermo-physical properties in an HTP manner using Thermo-Calc's Python API




a)




b)




c)

**Fig. 11 | Confusion matrix for Ni-Ti alloys printability map evaluation.** For the three NiTi alloys, the Confusion matrix is plotted the optimal combination of criteria, LOF2-KH1-Ball1 for **a**) $Ni_{50.3}Ti_{49.7}$ **b**) $Ni_{50.8}Ti_{49.2}$and **c**) $Ni_{51.2}Ti_{48.8}$, where 1 indicates the presence of the respective label in the printability maps and 0 indicates its absence.

can be found in ref. 53. Apart from CALPHAD-based material properties, the rule-of-mixtures (ROM) is used to calculate average boiling temperatures and molecular weights. These material properties are queried in an HTP manner and used in the expressions found in Sections "Calculating the Melt Pool Geometry" and "Evaluating the predicted printability maps." Table 5 summarizes all the properties queried from Thermo-Calc and ROM. Additional properties, such as effective heat capacity, can be derived using the properties listed in Table 5.

### Calculating the melt pool geometry

Utilizing the material properties from CALPHAD and ROM models, the melt pool dimensions of single-track prints are estimated with the Eagar-Tsai model (E-T)[54]. The E-T model was originally developed for predicting quasi-steady-state temperature fields during certain welding processes but is now widely applied in metal AM processes[55–57]. The E-T model predicts isotherm shapes around a moving heat source in a semi-infinite medium based on thermo-physical properties and processing conditions, such as laser power, scan speed, and beam diameter. The E-T model requires the following inputs: thermal conductivity, thermal diffusivity, density, specific heat, melting (liquidus) temperature, laser power, and scan speed.

The E-T model does not accept temperature-dependent material properties. Therefore, the thermal conductivity ($\kappa$) and density ($\rho$) are evaluated at the liquidus temperature of the alloy. The effective specific heat was used for specific heat ($c_p$) as it accurately reflects a material's heat absorption ability during the entire heating and melting process in AM. Laser absorptivity, $\eta$, was estimated using Drude's theory with Equation (5), where $\rho_0$ is the electrical resistivity ($\Omega$ m) and $\lambda$ is the laser wavelength (m)[15,51,58]. The electrical resistivity is queried at room temperature.

$$\eta = 0.365\sqrt{\frac{\rho_0}{\lambda}} \tag{5}$$

The E-T analytical model addresses the heat conduction equation by considering a Gaussian-distributed heat source moving over a semi-infinite plate. Melt pool dimensions (length, width, and depth) are determined by identifying where the temperature distribution has reached the alloy's melting temperature. Key assumptions in the E-T model involve neglecting temperature-dependent thermal properties, neglecting latent heat of melting and evaporation, and assuming a quasi-steady state. Another limitation of the E-T model occurs when the melt pool transitions from conduction to keyhole mode, as it underestimates the melt pool depth due to missing keyholing mode physics (e.g., deepening of the melt pool from recoil pressure caused by metal evaporation)[15].

To address this missing physics in the E-T model, the melt pool depth is estimated using the Gladush and Smurov (G-S) model[59]. Developed for welding processes, the G-S model is applicable to L-PBF and is based on the thermal balance and mechanical equilibrium of a gas-vapor keyhole. Zhu et al.[15] and Honarmandi et al.[60] demonstrated that the G-S model provides a more accurate melt pool depth prediction during keyholing compared to the E-T model. The G-S melt pool depth, $D$, is calculated using processing and material properties as shown in Eq. (6), where $P$ is the laser power, $k$ is the thermal conductivity, $r_0$ is the beam radius, and $\alpha$ represents the thermal diffusivity. Thermo-physical properties are calculated at the liquidus temperature of the alloy. The corrected depth from Eq. (6) is used to evaluate balling and lack-of-fusion criteria, defining regions with different macroscopic printing defects.

$$D = \frac{AP}{2\pi k T_{boiling}} \ln\left(\frac{r_0 + \frac{\alpha}{v}}{r_0}\right) \tag{6}$$

However, the authors realize that even with the correction with the G-S model, the E-T and G-S models ignore fluid flow effects, which are crucial during the additive manufacturing process. Fluid flow, driven by factors such as Marangoni convection, significantly influences the melt pool behavior and defect formation. For instance, variations in surface tension due to temperature gradients can cause fluid movement within the melt pool, affecting its shape and solidification dynamics. Additionally, the E-T and G-S models do not consider the temperature dependence of material properties, which can vary significantly under different processing conditions. Moreover, the applicability of these analytical models under high-speed scanning conditions in L-PBF has not been thoroughly discussed. Fast scanning speeds can exacerbate the thermal gradients and fluid dynamics

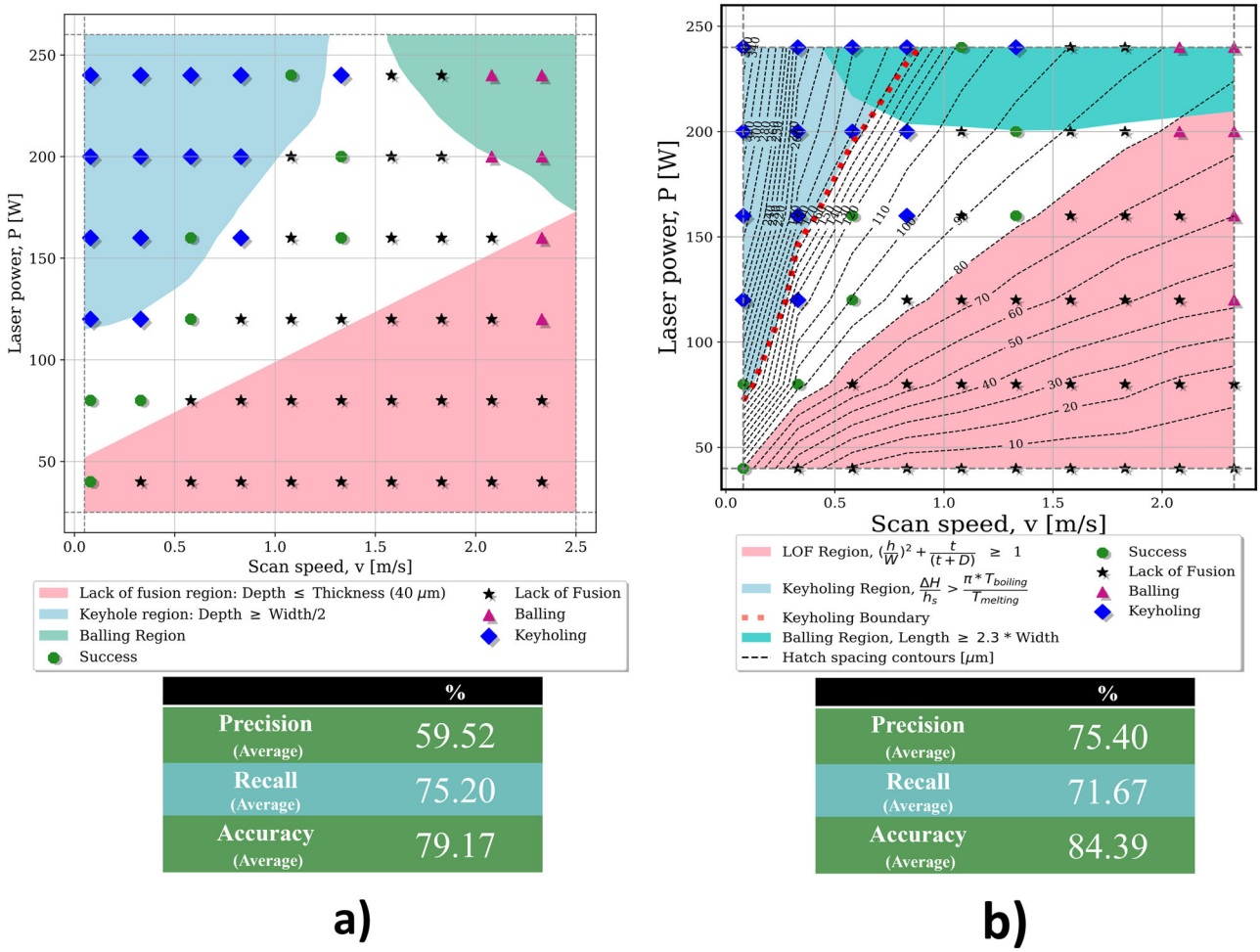



**Fig. 12 | Comparison between calibrated printability map and current framework printability map. a)** The printability map for $Ni_{50.3}Ti_{49.7}$ constructed using a calibrated E-T model, **b)** The fully computational printability map for the same alloy generated by the framework presented in this work. For a 10 by 10 design grid of laser power and scan speed, the time required to perform experiments and produce the calibrated printability map in part a) is around 10 days, while it takes just around 2-3 hours to produce the printability map in part **b)** using our framework. W: Melt pool width. h = Hatch spacing. t = Powder layer thickness. D = Melt pool depth. $\Delta H$: specific enthalpy. $T_{boiling}$: Boiling temperature. $T_{melting}$: melting temperature. $h_s$: Specific enthalpy at melting. Green circles: Experimentally observed defect-free single tracks. Salmon: Area that represents the predicted lack-of-fusion region. Blue: area that represents the predicted keyholing region. Teal: area that represents the predicted balling region. Black star: experimentally observed lack-of-fusion. Magenta triangles: Experimentally observed balling. Blue diamond: Experimentally observed keyholing. Red dashed line: Keyholing boundary. Black dashed lines: Maximum hatch spacing required for sufficient overlap in a fully built part.

within the melt pool, leading to defects such as keyholing and balling. The E-T and G-S models, while efficient, do not account for these complex interactions comprehensively. Future work should focus on integrating more detailed fluid flow simulations and temperature-dependent material properties to improve the predictive accuracy of these models. However, these models have been recognized to be acceptable methods to use for an initial outlook into predicting the melt pool morphology for L-PBF[20,61–63].

### Construction of printability maps

Upon predicting the melt pool profile, we evaluate criteria for keyholing, lack-of-fusion, and balling. The union of these criteria delineates areas in the processing space where defects occur, i.e., a printability map. However, several criteria exist for the same defect mode. Given the multiplicity of defect criteria, a key goal of this work is to benchmark these various criteria sets against available experimental data.

For lack-of-fusion, two criteria were evaluated (Equation (7) and Equation (8)), where $D$ is the melt pool depth, $W$ is the melt pool width, and $t$ is the powder layer thickness:

$$D \leq t \tag{7}$$

$$\left(\frac{h}{W}\right)^2 + \frac{t}{t+D} \geq 1 \tag{8}$$

Lack-of-fusion occurs when the melt pool does not penetrate deep enough into the powder layer to sufficiently bond with the substrate and/or preceding layers. This scenario can be described mathematically by Equation (7), where lack-of-fusion occurs when the melt pool depth is less than the powder layer thickness[2]. A second criterion, Equation (8), introduced by Zhu et al.[15], posits that excessive hatch spacing $h$ leads to inadequate binding of adjacent single-tracks, resulting in lack-of-fusion. If the hatch spacing $h$ surpasses the maximum hatch spacing $h_{max}$, insufficient overlap between melt tracks can induce porosity[4,15]. To mitigate this, the hatch spacing must be adjusted to ensure comprehensive bonding between adjacent tracks and previous layers. Maximum hatch spacing is calculated using Equation (9)[4]:

$$h_{max} = W\sqrt{1 - \frac{t}{t+D}} \tag{9}$$

Using Equation (9), it is possible to estimate the maximum hatch spacing that would still result in fully dense layers at different locations of the

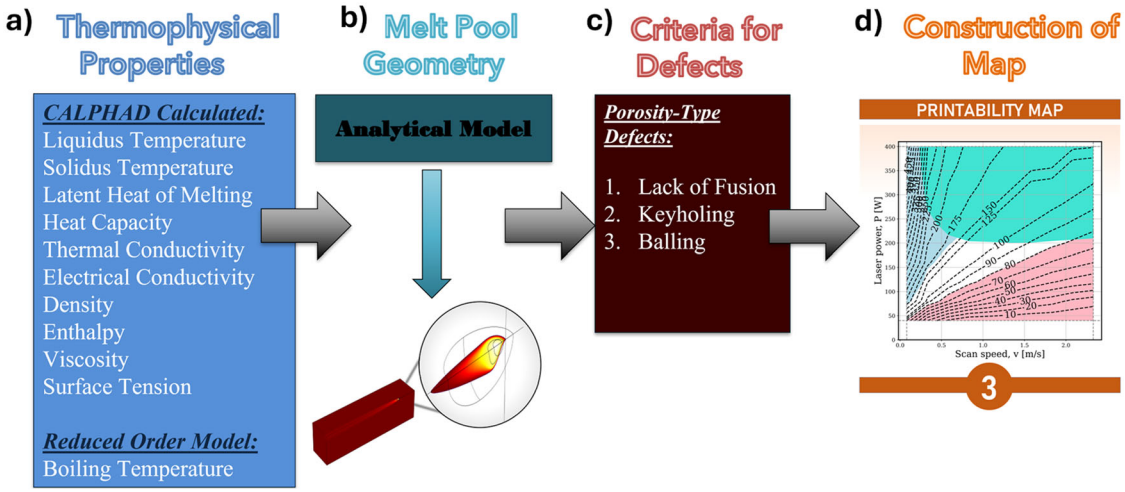



**Fig. 13 | Step to the framework to predict printability maps.** The introduced framework can be divided into four distinct stages: **a)** thermophysical property calculation for the material system using a CALPHAD-based approach or reduced-order models, **b)** melt pool geometry calculation, **c)** defect criteria selection, and **d)** printability map construction.

**Table 5 | Thermophysical properties queried in this work.**

| Property | Symbol | Source/Equation |
|---|---|---|
| Liquidus | $T_L$ | Equilibrium Module |
| Solidus | $T_S$ | Equilibrium Module |
| Boiling Temperature | $T_B$ | ROM |
| Molecular Weight | $MW$ | ROM |
| Electrical Resistivity (RT) | $\rho_0$ | Property Module |
| Thermal Diffusivity ($T_L$) | $\alpha$ | Property Module |
| Thermal Conductivity ($T_L$) | $\kappa$ | Property Module |
| Density ($T_L$) | $\rho$ | Property Module |
| Enthalpy (RT, $T_S$, $T_L$) | $H_{RT,S,L}$ | Equilibrium Module |
| Solidification Range | $\Delta T$ | $T_L - T_S$ |
| Latent Heat of Fusion | $L_f$ | $H_L - H_S$ |
| Latent Heat of Vaporization | $H_V$ | $10RT_L$ |
| Effective Heat Capacity | $C_p$ | $\frac{H_L - H_{RT}}{T_L - T_{RT}}$ |
| Absorbtivity | $\eta$ | $0.365\sqrt{\frac{\rho_0}{\lambda}}$ |

process map. These contours have also been plotted onto the printability maps, as shown in this paper.

Balling occurs when the melt pool fragments into droplets due to Plateau-Rayleigh capillary instability. Balling typically occurs at high laser power and scan speeds when the melt pool is elongated. Experimental comparisons across common AM alloys suggest a threshold of 2.3, as proposed by Zhang et al.[2] and Johnson et al.[3], shown in Equation (10). For single-tracks, Yadroitsev et al.[48] identified a boundary between stable and unstable melt pool zones, forming the second balling criterion, described in Equation (11).

$$\frac{L}{W} \geq 2.3 \tag{10}$$

$$\frac{\pi W}{L} < \sqrt{\frac{2}{3}} \tag{11}$$

Keyholing-induced pores form when rapid molten material evaporation creates vapor cavities propelled deeper into the melt pool by recoil pressure. This recoil pressure leads to a larger melt pool depth compared to the conduction mode. The collapse of these cavities leaves behind voids. Keyholing is typically observed at high laser power and low scan speeds within the processing space.

Johnson et al.[3] proposed a keyholing criterion based on the melt pool's depth-to-width ratio, deriving a threshold of 1.5 from experiments and geometric reasoning. Applying the same theory, we set a general threshold of 2.75 for additive manufacturing (AM) alloys as per Equation (12), while adjusting it to 2.0 for NiTi-based alloys based on Zhang et al.'s[2] experimental findings (Equation (13)). In this study, we retain these thresholds, but they can be varied to explore the printability of arbitrary compositions.

$$\frac{W}{D} \leq 2.75 \tag{12}$$

$$\frac{W}{D} \leq 2.0 \tag{13}$$

Keyholing onset can also be estimated through criteria considering material properties and processing parameters beyond the melt pool geometry. For instance, King et al.[20,62] identified a positive correlation between normalized enthalpy and melt pool depth, thereby linking it to keyholing. They derived a keyholing criterion (Equation (14)) that includes the specific enthalpy at melting, $h_s = H_L/MW$, where $H_L$ is the enthalpy at the liquidus, and $MW$ is the molecular weight of the alloy. Importantly, King et al.'s original formulation underestimates a metal's energy absorption from the laser due to using the solid phase heat capacity. Therefore, we replaced this with the *effective heat capacity*, accounting for the alloy's sensible and latent heat from room temperature up to the liquidus.

$$\frac{\Delta H}{h_s} = \frac{\eta P}{\pi h_s \sqrt{\alpha v r_0^3}} > \frac{\pi T_B}{T_L} \tag{14}$$

Gan et al.[64] recently formulated a universal keyholing criterion via dimensional analysis and a modified Buckingham-Pi theorem-based framework. This single metric defines the melt pool's conduction, transition, and keyhole modes. The dimensionless keyhole criterion, $Ke$ (Equation (15)), is a function of processing parameters (laser power $P$, scan speed $v$, beam radius $r_0$) and material properties (absorptivity $\eta$, liquidus temperature $T_{\text{liquidus}}$, substrate temperature $T_0$, density $\rho$, specific heat capacity $C_p$, and thermal diffusivity $\alpha$). Using a limited experimental dataset, Gan et al.

**Table 6 | The equations and labels for each respective defect criterion**

| Defect Type | Label | Equation |
|---|---|---|
| Lack-of-Fusion | LOF1 | $D \le t$ |
| | LOF2 | $\left(\frac{h}{W}\right)^2 + \frac{t}{t+D} \ge 1$ |
| Balling | Ball1 | $\frac{L}{W} \ge 2.3$ |
| | Ball2 | $\frac{\pi W}{L} < \sqrt{\frac{2}{3}}$ |
| Keyholing | KH1 | $\frac{W}{D} \le 2.75$ (for general AM alloys) |
| | | $\frac{W}{D} \le 2.0$ (for NiTi-based alloys) |
| | KH2 | $\frac{\Delta H}{h_s} = \frac{AP}{\pi h_s \sqrt{\alpha v a^3}} > \frac{\pi T_{boiling}}{T_{liquidus}}$ |
| | KH3 | $Ke = \frac{\eta P}{(T_{liquidus} - T_0)\pi\rho C_p \sqrt{\alpha v r_0^3}} > 6$ |

A total of 12 printability maps can be generated with every combination of these defect criteria. Furthermore, 3 of these combinations were used to generate the printability maps in Fig. 9.

identified $Ke > 6.0$ as the threshold for keyholing.

$$Ke = \frac{\eta P}{(T_L - T_0)\pi\rho C_p \sqrt{\alpha v r_0^3}} > 6 \tag{15}$$

In this study, we evaluate 2 criteria for lack-of-fusion, 2 criteria for balling, and 3 criteria for keyholing. These criteria can be combined to yield 12 distinct printability maps. The various criteria are summarized in Table 6.

In summary, by estimating melt pool dimensions as a function of power and velocity, printability maps can be generated. For a particular power-velocity pair, the melt pool dimensions are predicted and fed into various defect criteria. These defect criteria are functions of melt pool geometry and, in some cases, alloy properties. If the power-velocity pair produces a melt pool classified as defective, the point in power-velocity space is colored according to its defect type (pink for lack of fusion, blue for keyholing, green for balling); otherwise, it remains uncolored. This process is repeated across the power-velocity space until satisfactory defect classification boundaries are established.

The transitions between different printing regimes are identified by the boundary lines on the printability map. In this work, we generate printability maps and predict the decision boundaries for various defects in $P - v$ space. We then check if the experimental data is correctly labeled as defective or not. If the data point falls on the boundary line for the specified defect, it is considered within the predicted defect region.

Because there are 12 possible combinations of defect criteria, 12 possible printability maps are explored in this work. The accuracy of each of these 12 maps and the potential to fuse information among them are explored in the following section.

### Evaluating the predicted printability maps

Determining the accuracy of these computed printability maps can be treated as a classification problem. To evaluate the performance of the 4 decision boundary types in the 12 printability maps, we use common performance metrics for binary classifiers: precision, recall, and accuracy. Precision (Eqn. (16)) measures the model's reliability in predicting the onset of a given defect (or the absence of all defects in the case of the printable region), while Recall (Eqn. (17)) assesses the model's ability to identify all process conditions leading to a specific defect. Accuracy (Eqn. (18)) provides an overall performance measure, combining precision and recall.

$$\text{Precision} = \frac{TA}{(TA + FA)} \tag{16}$$

$$\text{Recall} = \frac{TA}{(TA + FN)} \tag{17}$$

$$\text{Accuracy} = \frac{TA + TN}{TA + FA + TN + FN} \tag{18}$$

In the equations above, TA and N are the number of points in the positive and negative classes, and TA, TN, FA, and FN denote the number of true positives, true negatives, false positives, and false negatives, respectively.

We utilized a chemically diverse dataset to evaluate the performance of each of the 12 criteria sets. Table 3 lists the amount of single-track data available for five commonly printed alloys used to benchmark our framework. Each predicted defect class was evaluated using precision, recall, and accuracy metrics. However, the prevalence of the three defect modes is not necessarily equal, leading to class imbalance. To address this, we computed the weighted average of the performance metrics, considering the prevalence of each defect type within the experimental dataset. For instance, if 100 single-track experiments are reported for a given alloy and 25 points are classified as 'lack-of-fusion,' then the contribution of the 'lack-of-fusion' model to the overall performance metric would be 25%. While we computed precision, recall, and accuracy, the printability map was primarily evaluated in terms of weighted average accuracy.

The accuracy was calculated by considering a binary decision problem. The experimental points are overlaid on top of the generated printability maps, and the problem is sorted into four binary decision problems for each data point. We separate the problem into four decisions:

1. Lack of fusion
2. Keyholing
3. Balling
4. Success or Defect-Free

The class or positive class is the defect being evaluated, and the negative class is the absence of the defect at the particular power-velocity point for the data point. The accuracy formula in Eqn. (18) is applied for each class. However, to avoid bias from one defect or class affecting the accuracy of the printability map, we use a weighted accuracy. The weighted accuracy is calculated as follows:

$$p_i = \frac{\text{number of data points in class of defect}}{\text{number of total data points used in validation of alloy}} \tag{19}$$

where $p_i$ is the percentage of the data points experimentally classified as the said defect class. The weighted average accuracy is then calculated as:

$$\text{Accuracy}_{\text{weighted}}(\%) = \left(\sum_{i=1}^{n} p_i \times x_i\right) \times 100 \tag{20}$$

where $i$ represents the four classes we have defined, and $x_i$ is the accuracy value calculated using Eqn. (18). For an arbitrary alloy, "Alloy 1," suppose we have 100 experimental validation points, with 25 points each for LOF, balling, keyholing, and defect-free. Thus, $p_L = p_K = p_B = p_D = 0.25$ for each class, and after overlaying these points on the printability map, the accuracy values are: $x_L = 85\%$ for LOF, $x_K = 90\%$ for keyholing, $x_B = 75\%$ for balling, and $x_D = 80\%$ for defect-free. By employing the weighted accuracy equation, the average accuracy for Alloy 1 is determined to be 83%.

### Inclusion and ethics statement

Our study on a computational framework to predict printability maps for L-PBF manufactured alloys adheres to ethical standards, ensuring transparency in research methodologies and data handling. We are committed to promoting diversity and inclusivity in computational materials science, fostering an environment that values equity and accessibility in our research endeavors.

## Data availability

The primary data supporting this study's conclusions are provided within the manuscript. Additional datasets can be accessed upon request from the corresponding author.

## Code availability

The underlying code for this study is not publicly available but may be made available on reasonable request from the corresponding author.

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## Author contributions

S.S. led the creation of the manuscript, with critical contributions to various sections by B.V. and P.H. P.M. and D.S. provided support in compiling the code. I.K., A.E., and R.A. assisted in conceptualizing and guiding both the manuscript and the concepts presented. All authors participated in editing and reviewing the manuscript.

## Competing interests

The authors declare no competing interests.

## Additional information

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